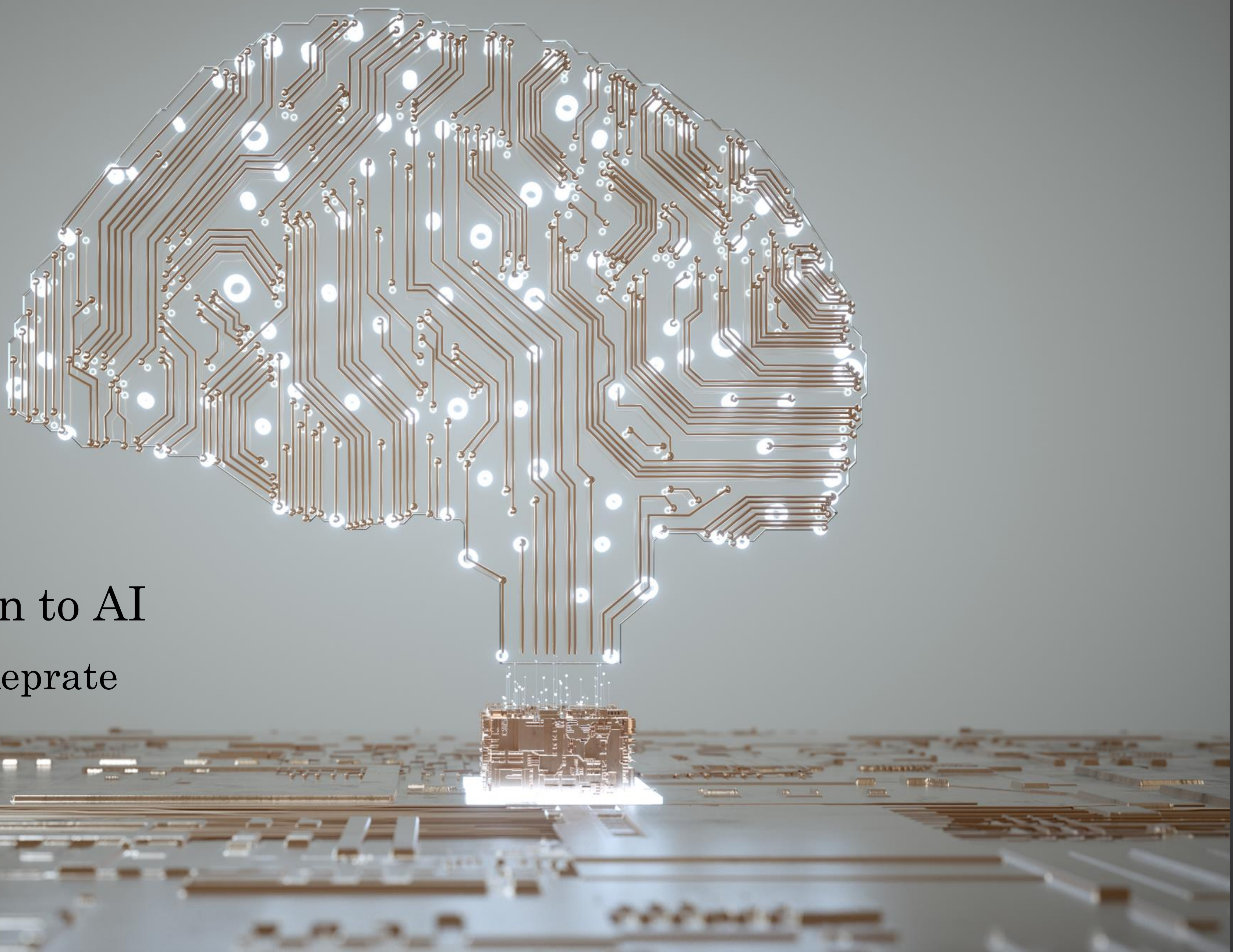
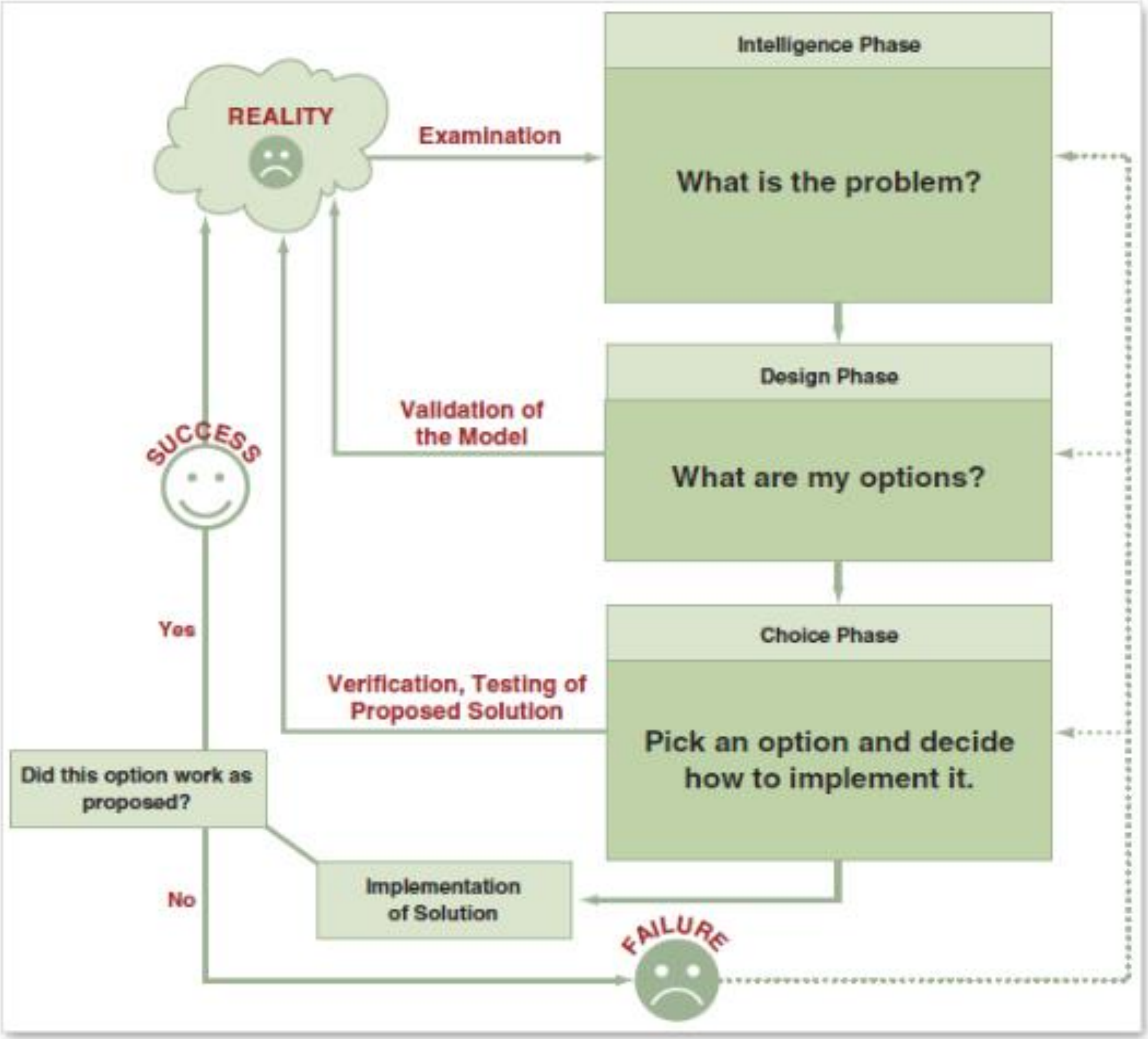
ML & NLP: PGR210



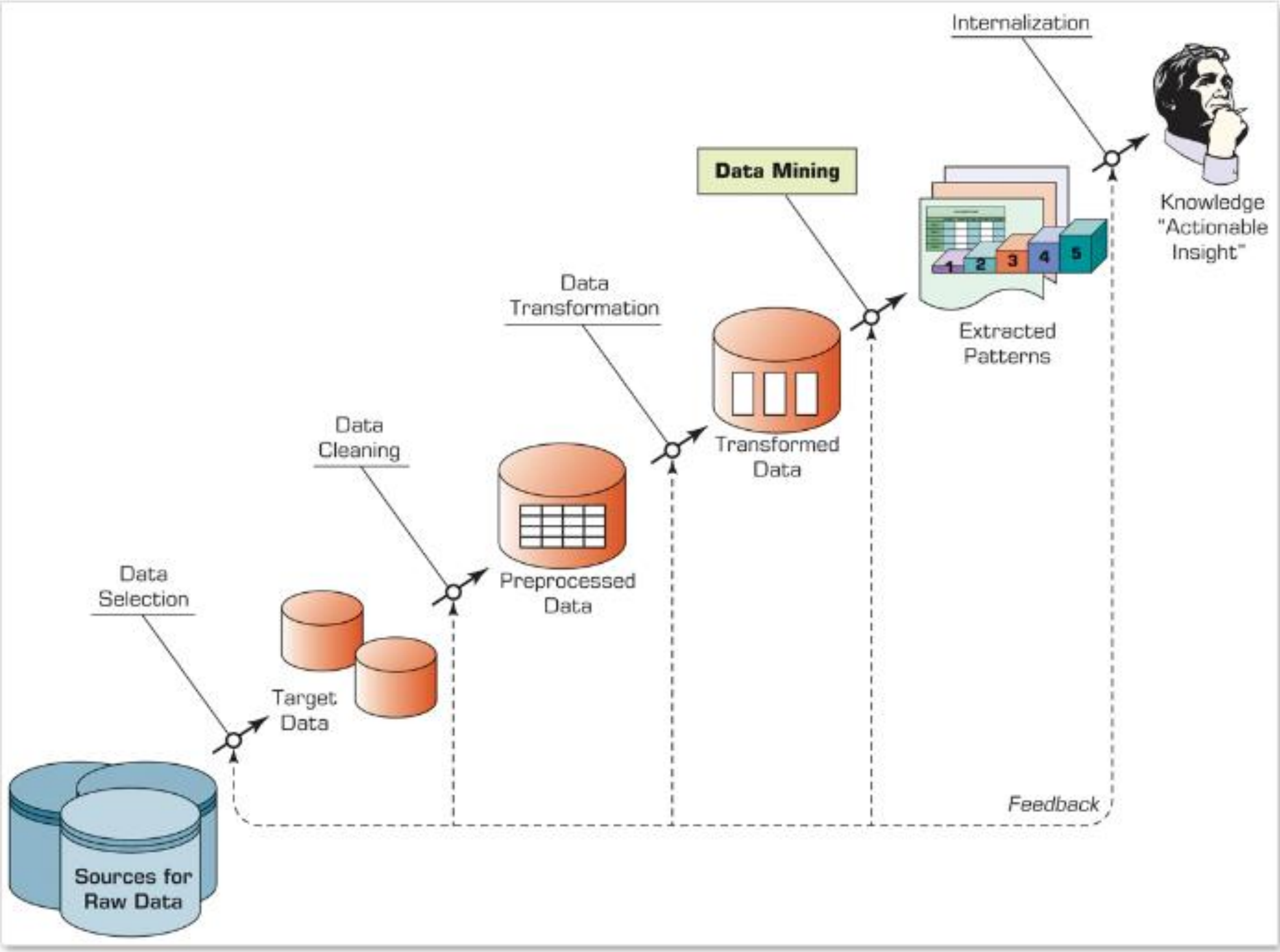
Introduction to AI

Dr. Arvind Keprate

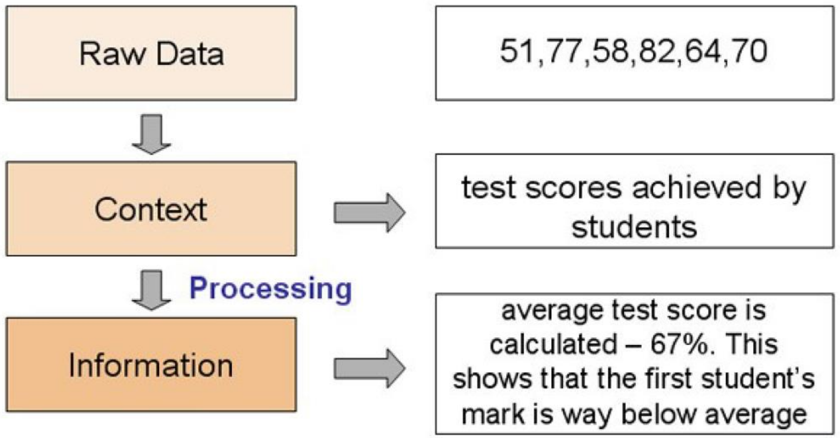


Decision Making

Knowledge Discovery in Databases (KDD) Process



Why Should We Analyze Data?



Data

We started off with the raw data which was 51, 77, 58, 82, 64, 70.

At this stage, those numbers could have been anything from street numbers to chart positions of certain records.

Context

The raw data is given a context. We are told that the numbers are test scores which were achieved by a group of students. At last they start to make some sense.

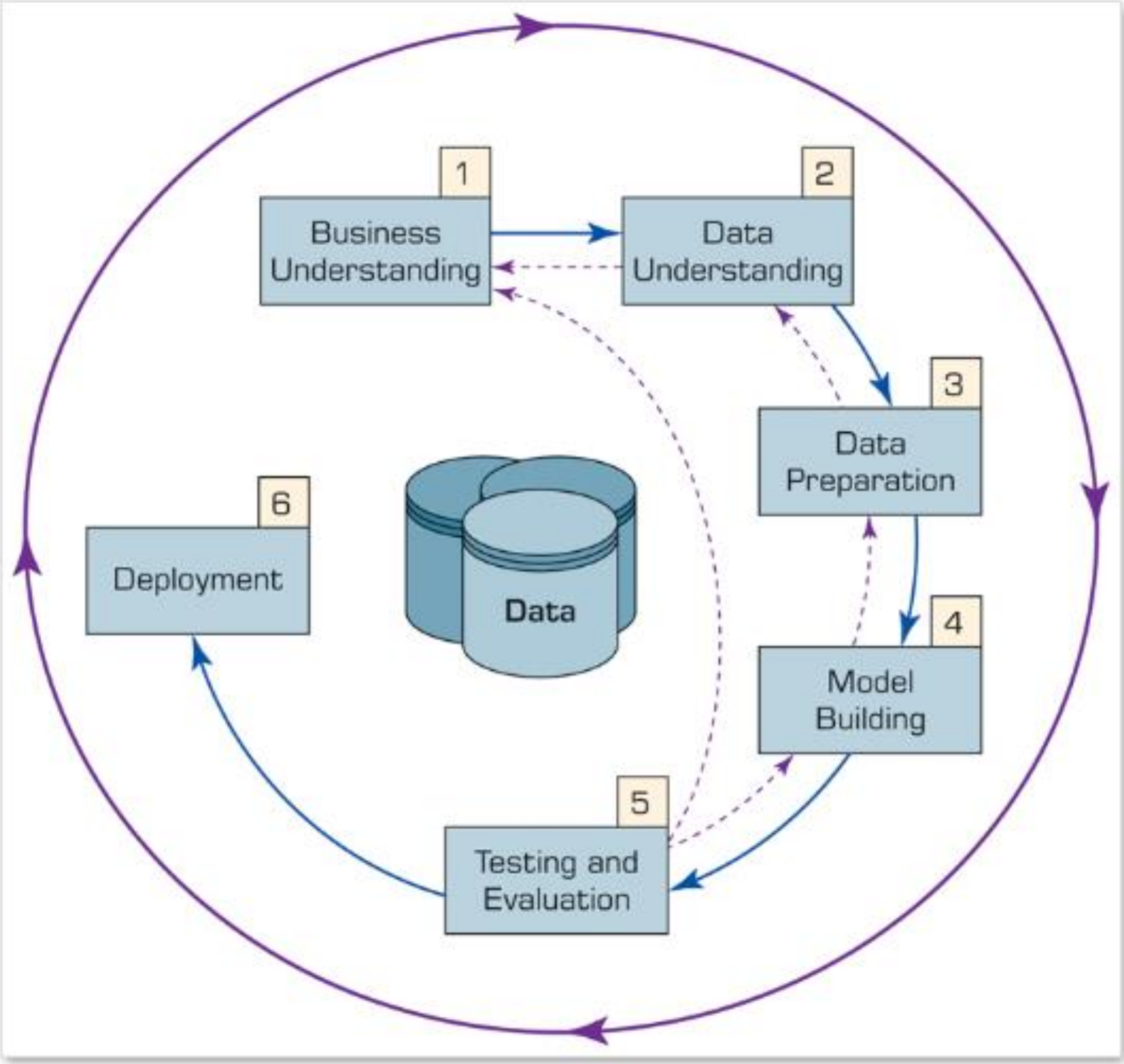
However even with a context, the data still needs to be processed in order to turn it into information.

Processing

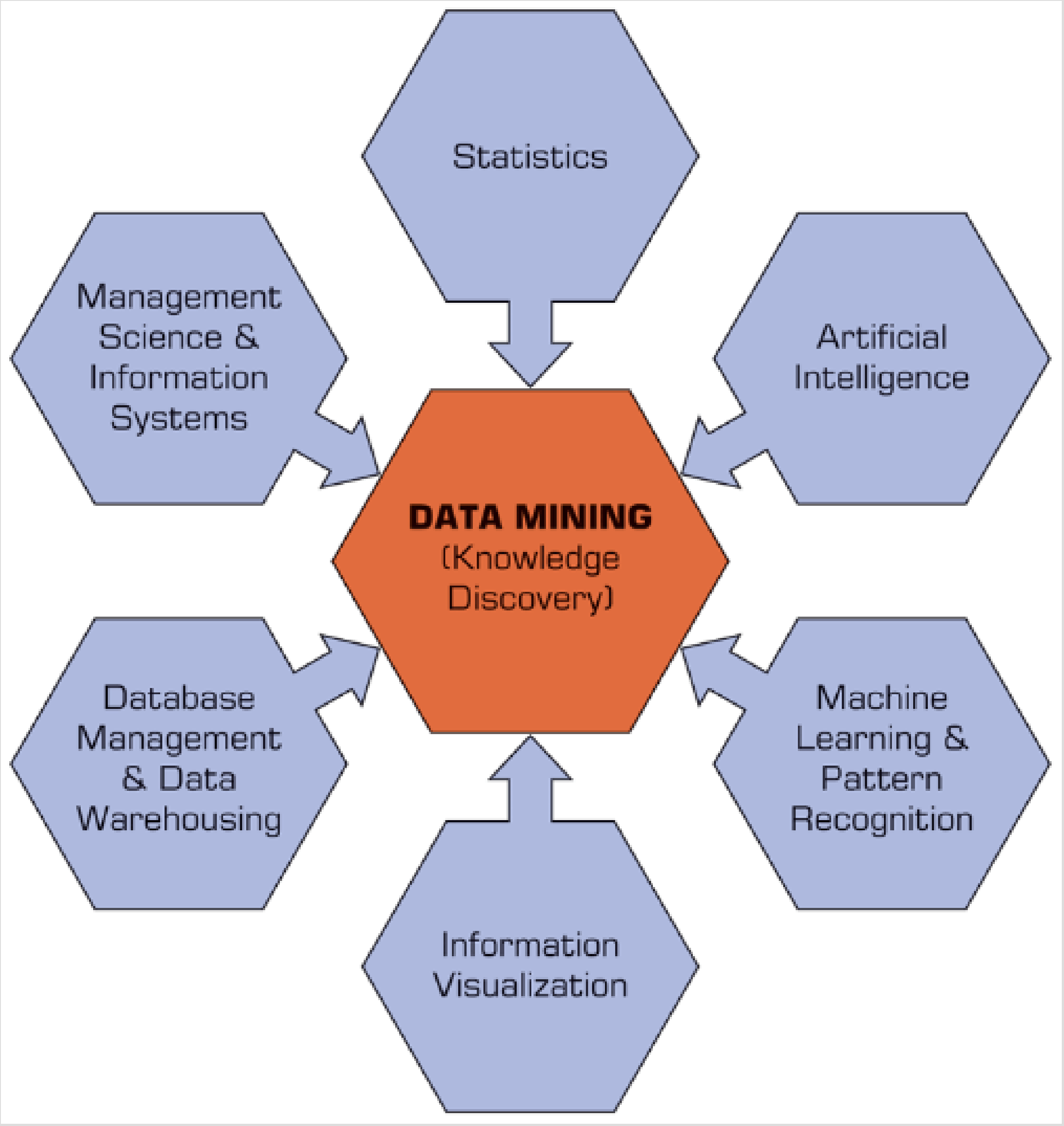
The processing in this case might be to calculate the average from the set of scores. The average is 67 so we can now tell that student 1 did particularly badly in the test because they were so far below the average mark.

Link

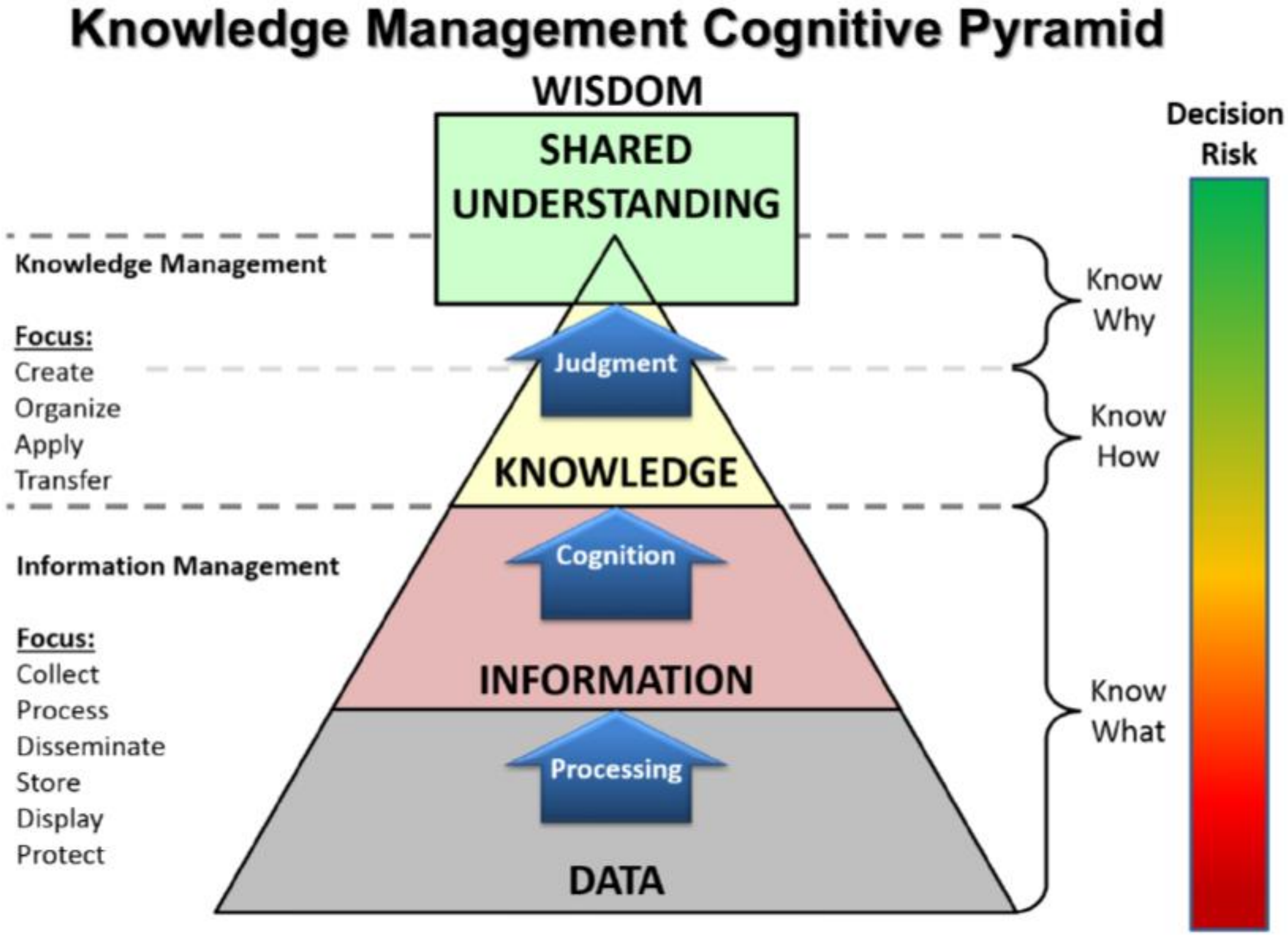
CRISP Data Mining Process



Data Mining


Statistics
Management Science & Information Systems
Artificial Intelligence
DATA MINING (Knowledge Discovery)
Database Management & Data Warehousing
Machine Learning & Pattern Recognition
Information Visualization

Data Information Knowledge Wisdom (DIKW) Pyramid



DIKW Adaptation in use within the US Army KM Community of Practice by way of
https://commons.wikimedia.org/wiki/File:KM_Pyramid_Adaptation.png

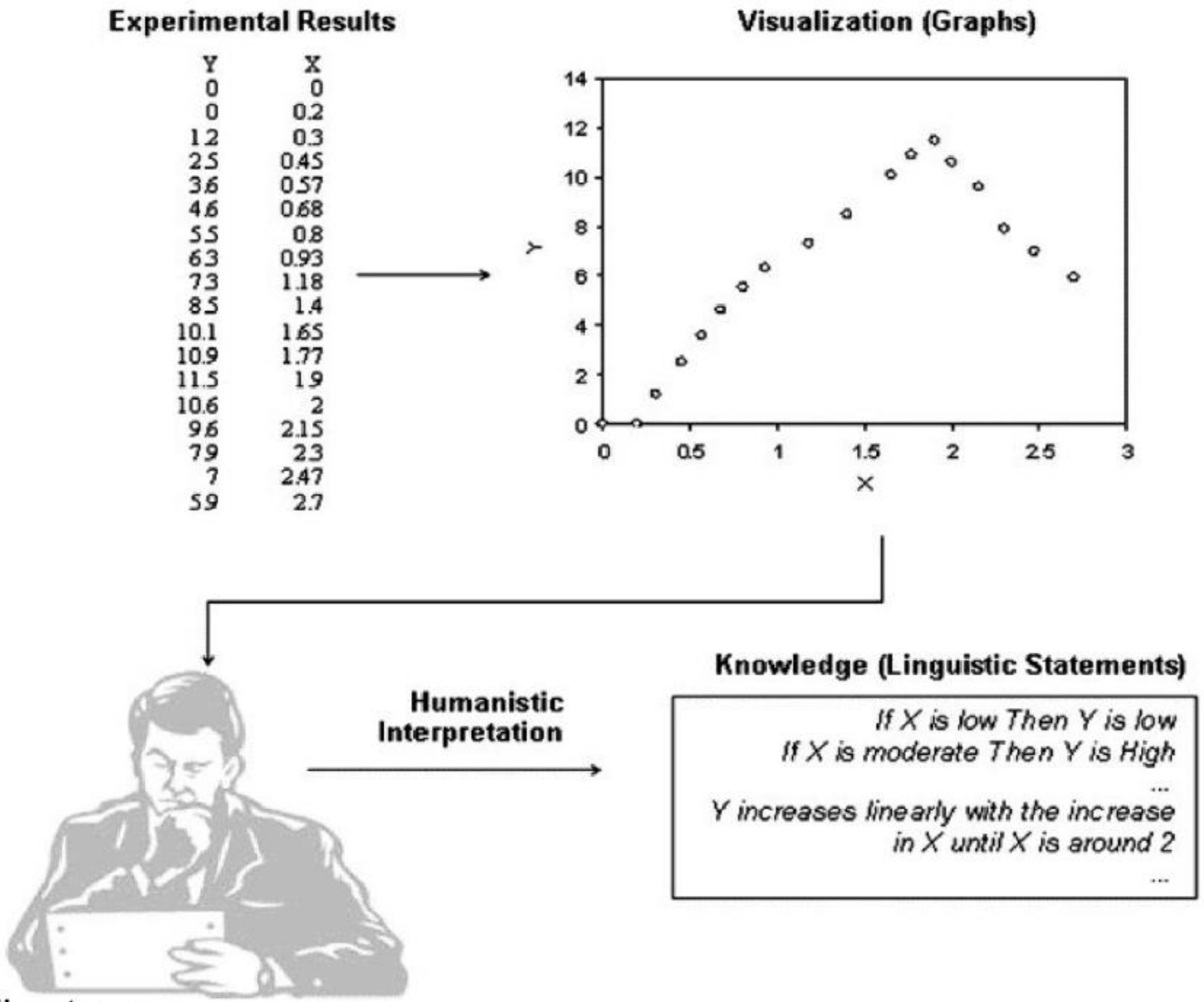
DIKW Progression

Fig 1 Knowledge Progression

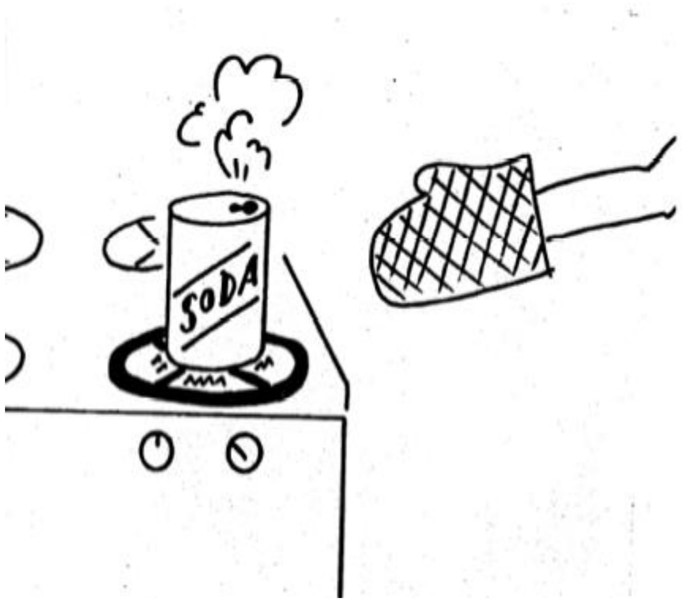
Fig. Knowledge Model

share.pdfonline.com

DIKW Example



sciencedirect.com

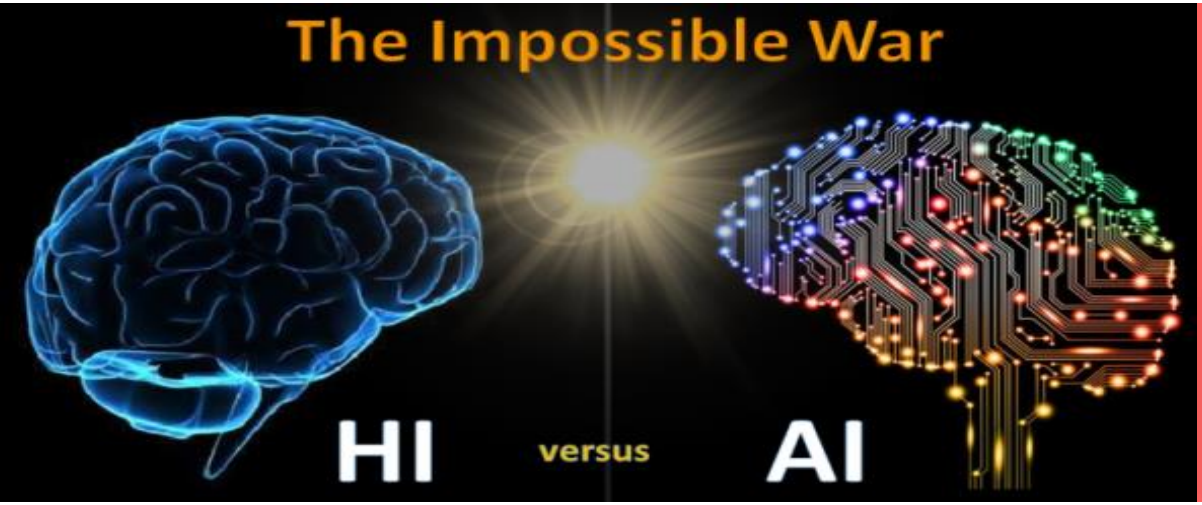


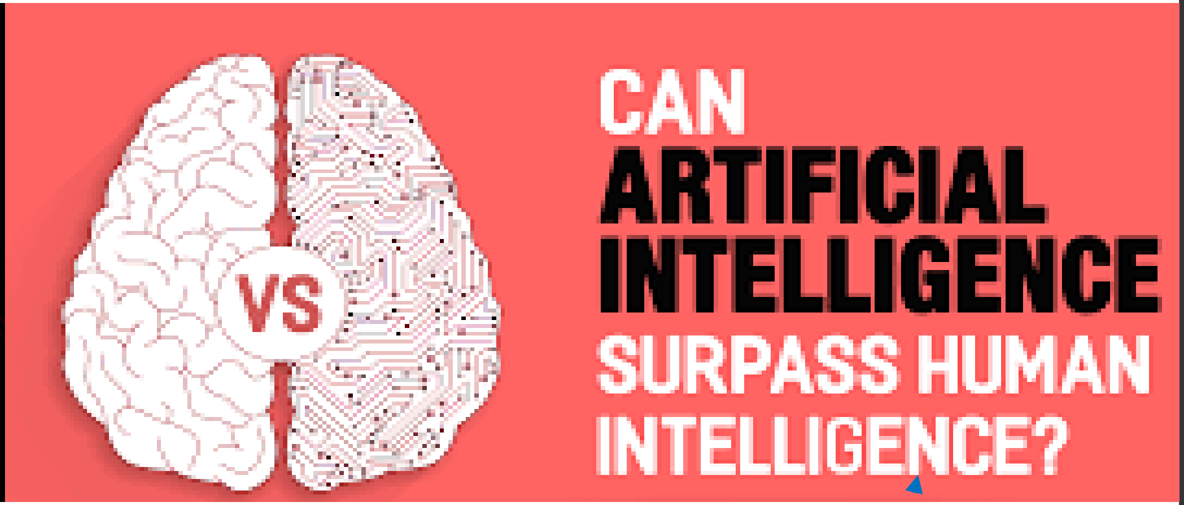
- Remembering your hand was burned the first time is not intelligence.

- Not touching the stove the second time is an example of intelligence, the ability to gain and apply knowledge and skills.

- Intelligence is the ability to take your memory then do something based on the details of the memory [1].

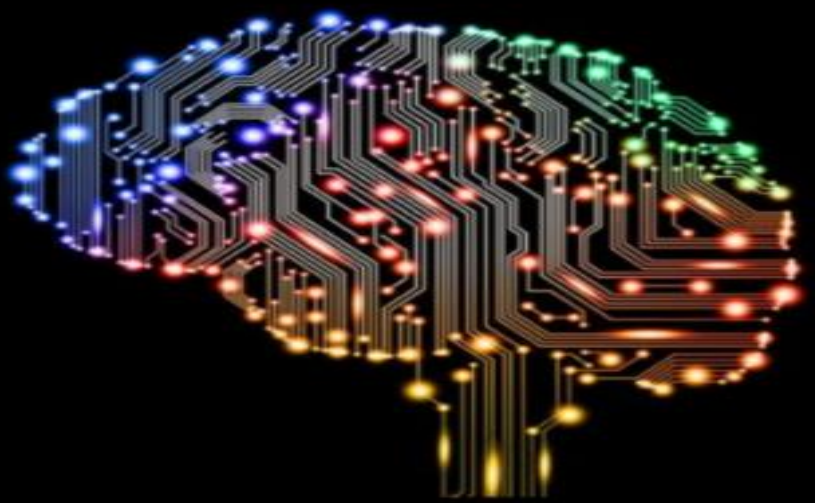
What is Artificial Intelligence (AI)?





- Human intelligence revolves around adapting to the environment using a combination of several cognitive processes.
- The field of Artificial Intelligence is a branch of computer science focusing on designing machines that can mimic human behavior [2].
- AI researchers are able to go as far as implementing Weak AI, but not the Strong AI. In fact, some believe that Strong AI is never possible due to the various differences between the human brain and a computer.

Artificial Intelligence vs. Human Intelligence



Machine Intelligence

- Fact-based and law-governed: data and algorithms
- Adaptive to environments with abundant data, more dimensions of variables, and greater predictability
- More accurate and effective at deterministic tasks

Human Intelligence

- Experience-based; observation, learning and culture
- Adaptive to environments with high uncertainty and low predictability
- More competent in judgement, creativity

Ref. [3]

AI History Timeline

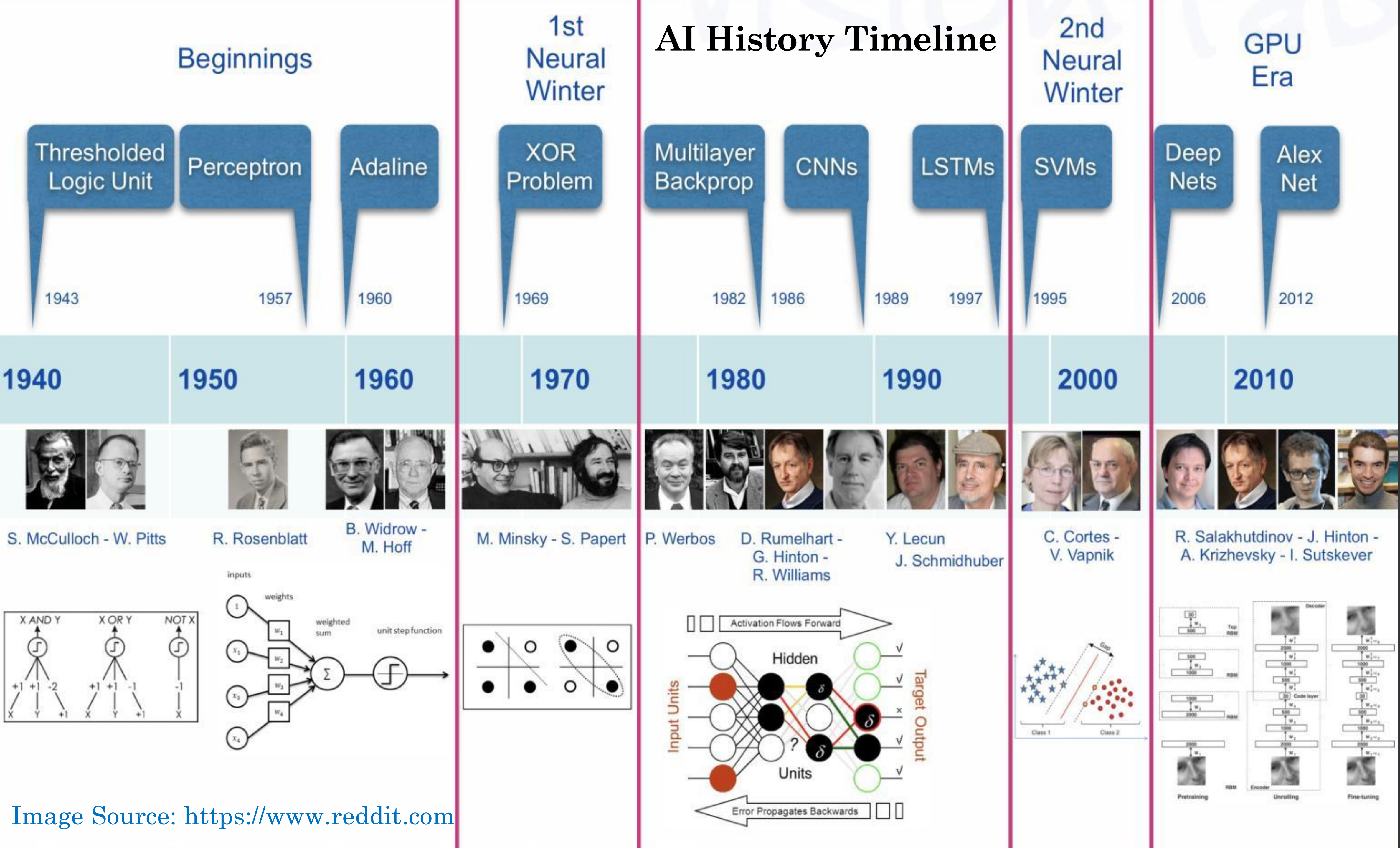
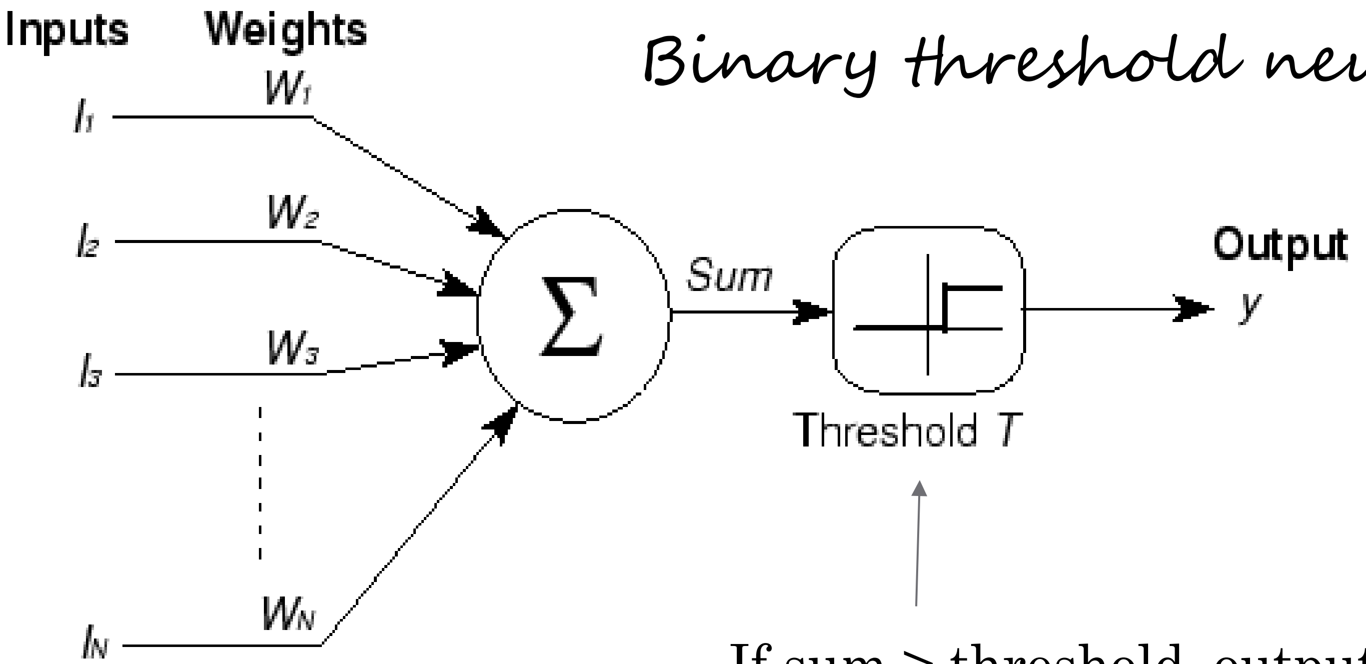


Image Source: https://www.reddit.com

AI History

McCullogh & Pitts - 1943

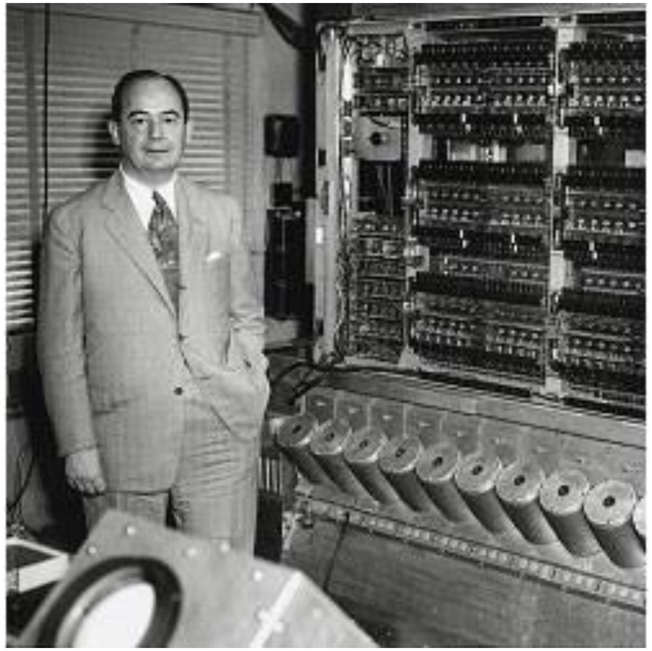
Binary threshold neuron



If sum > threshold, output 1
Output 0 otherwise

Weighted inputs

Von Neumann used this logic when designing the "universal computer"

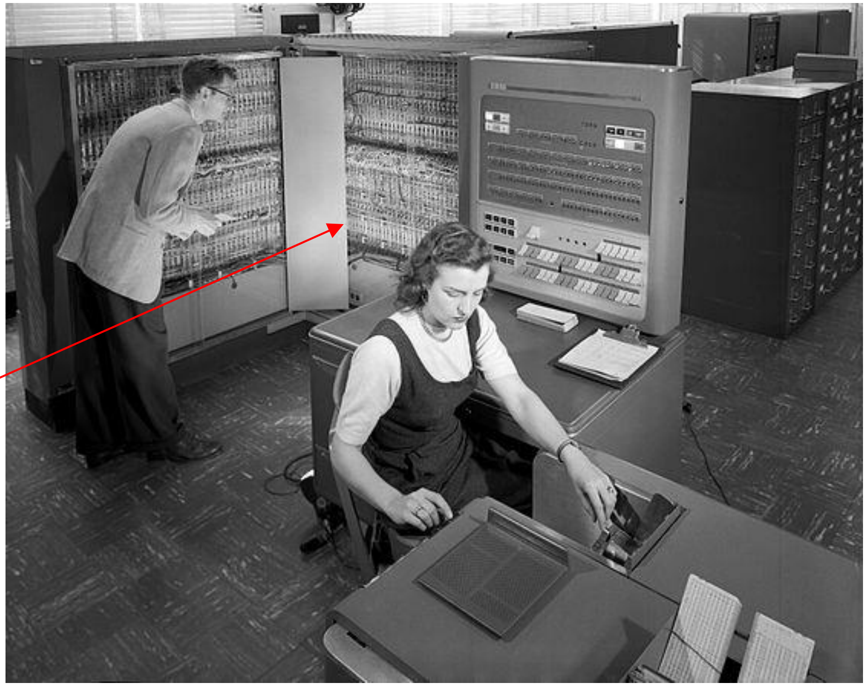


AI History

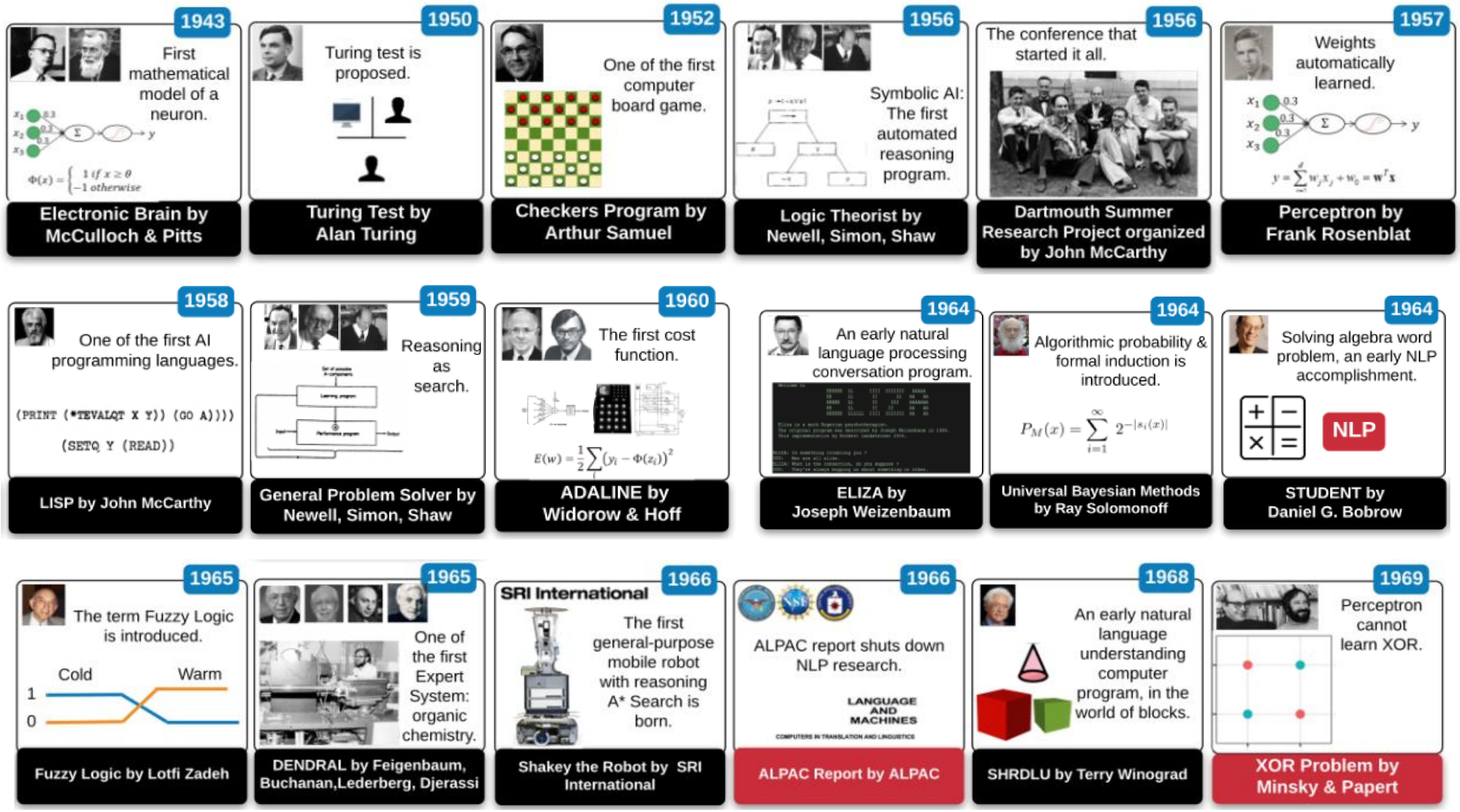
In the 1950s and 1960s, Principles of Neurodynamics were examined and Symbolic ML expanded

When I show these shapes to the camera

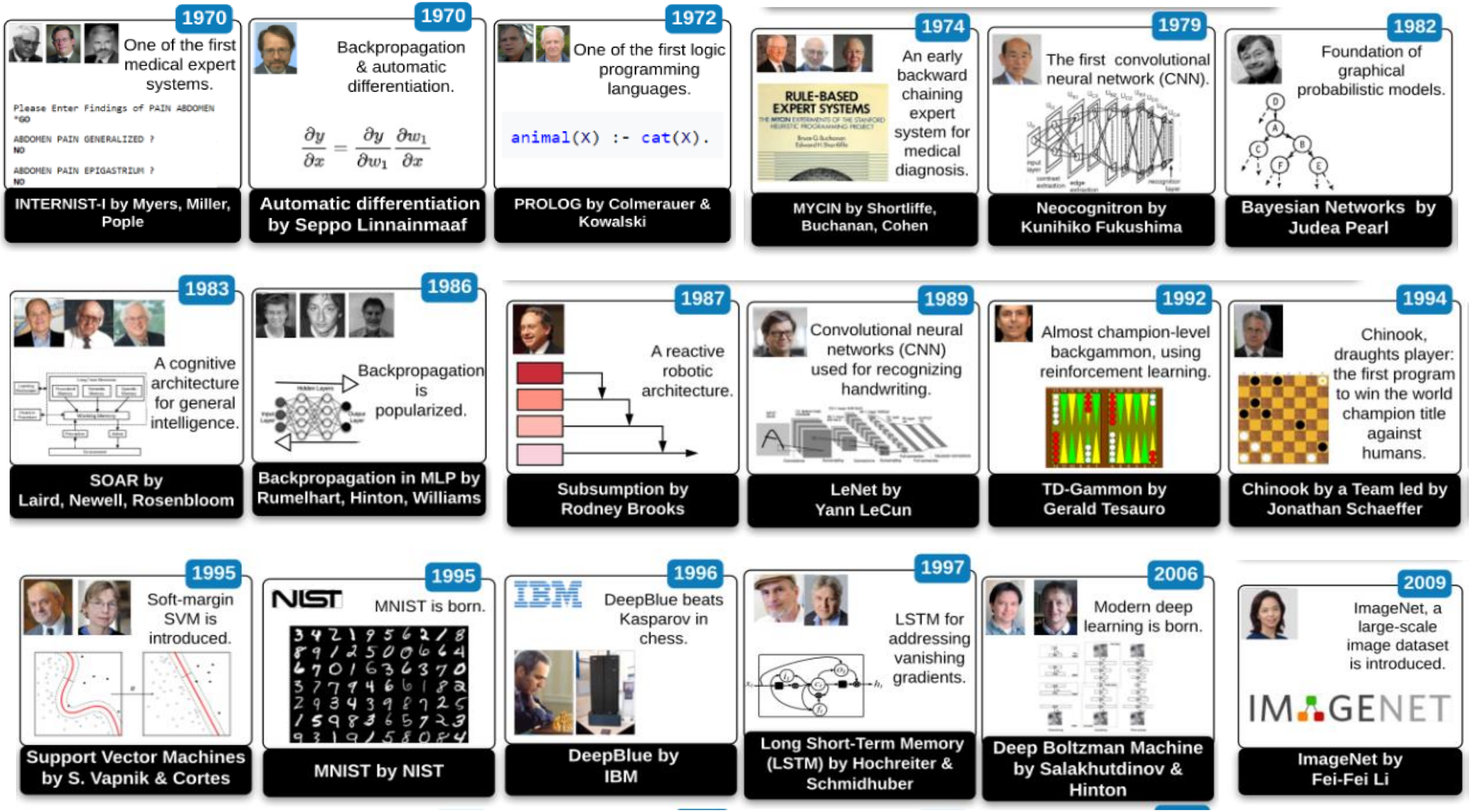
This IBM 704 computer can say “it’s a triangle”



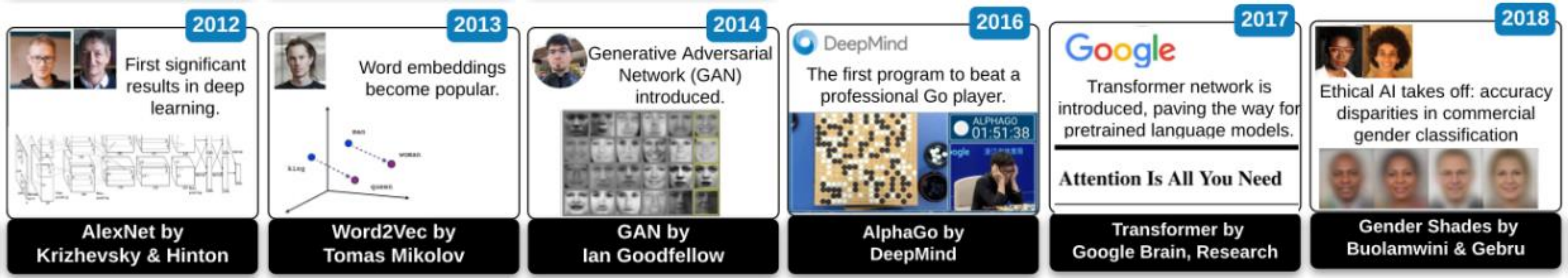
AI Evolution



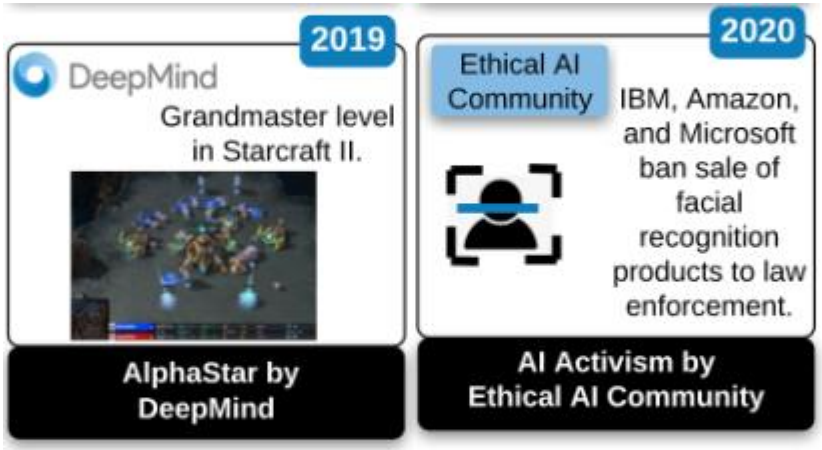
AI Evolution



AI Evolution

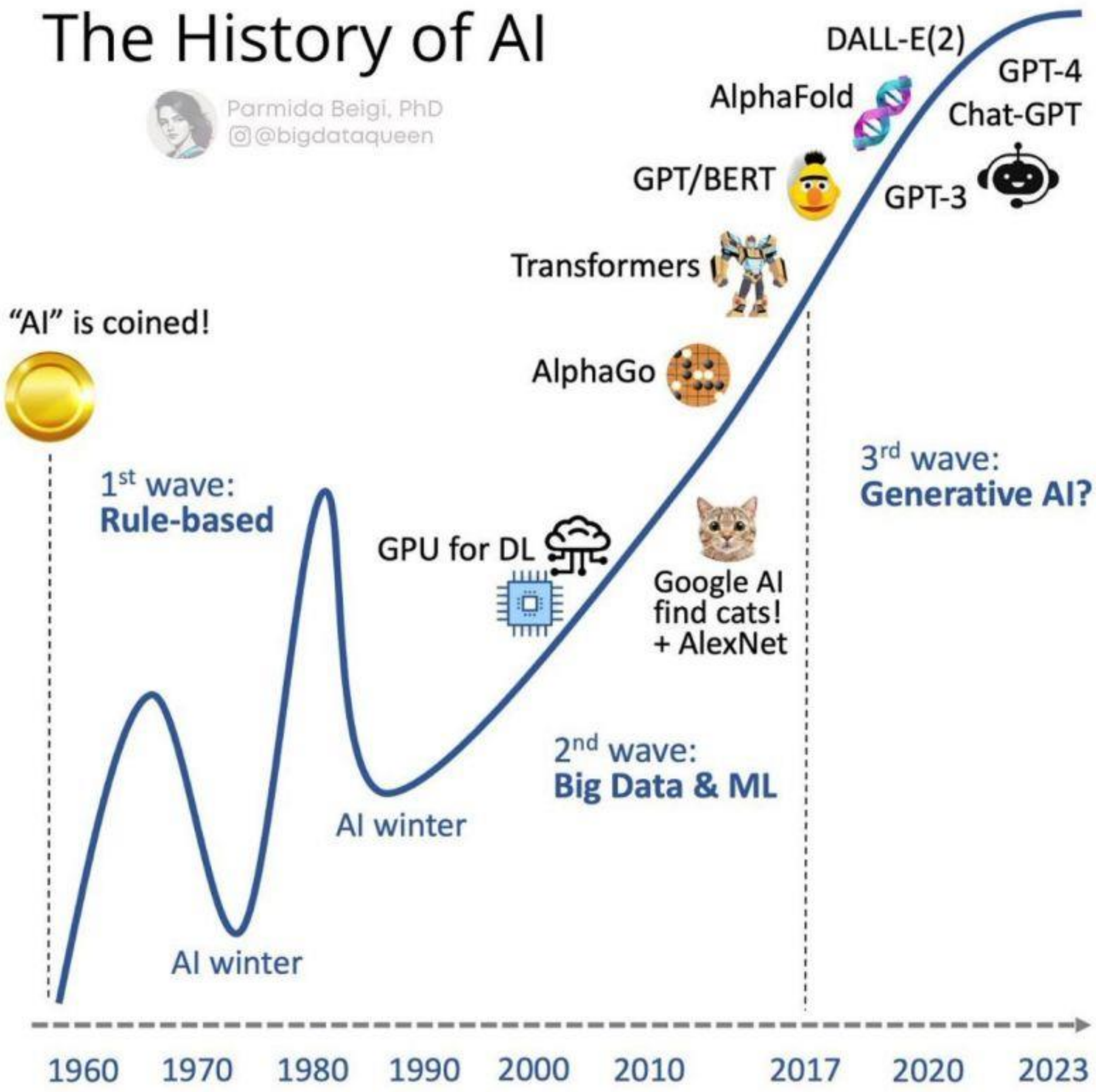


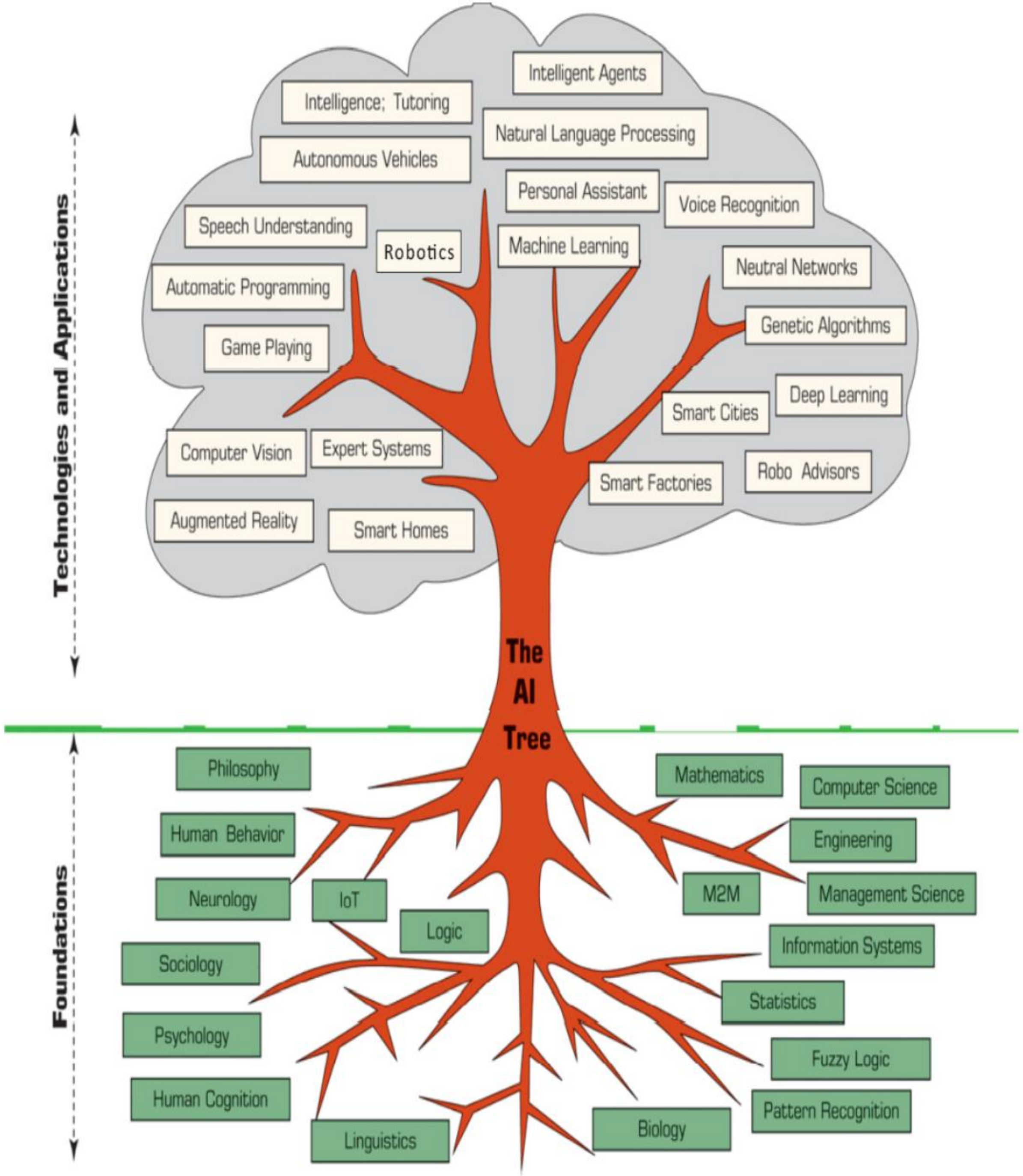
Large Language Models



Winter of Machine Learning

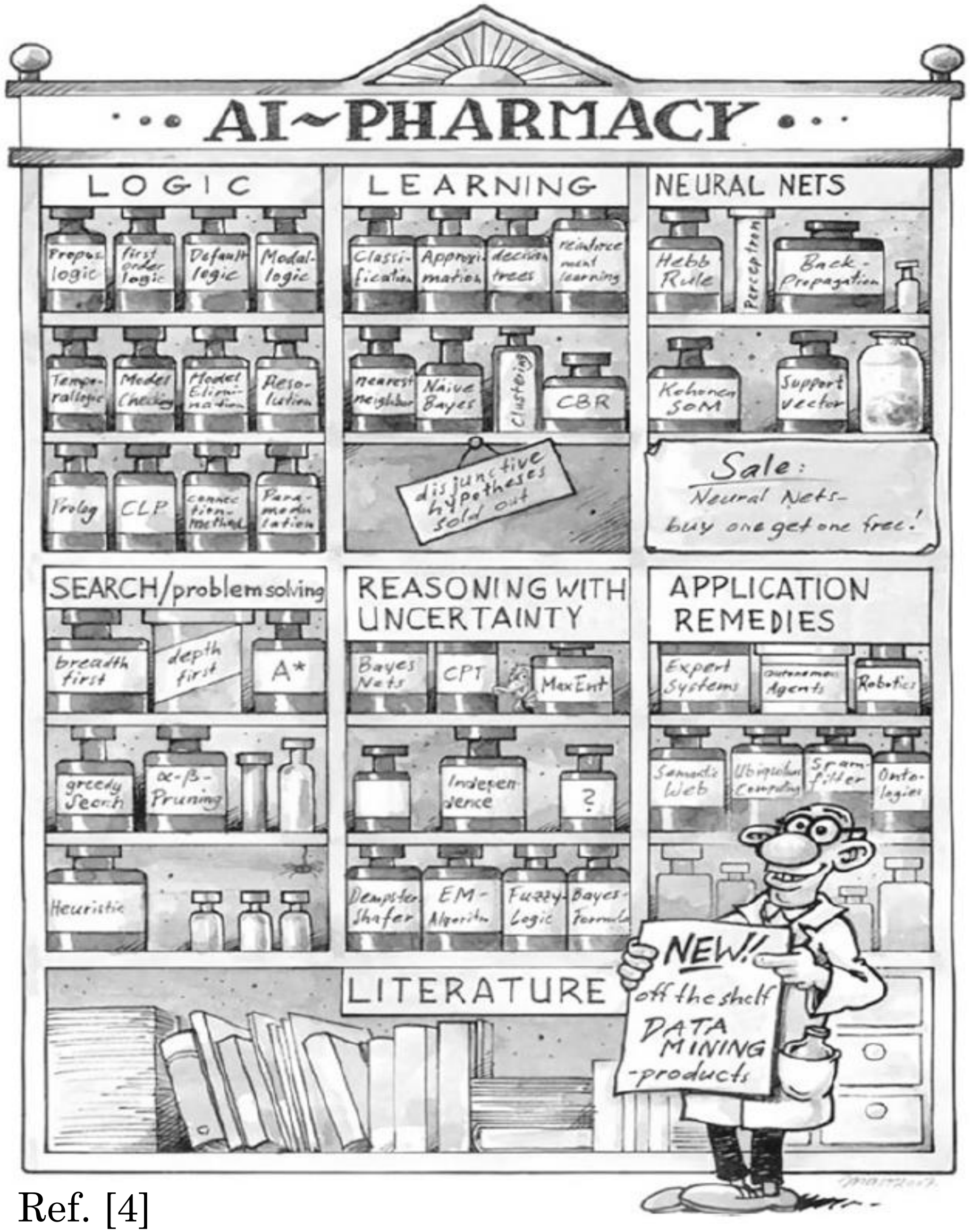
During the 1970s and 1980s, AI experienced what is known as the AI winter. Progress was slower than anticipated, and public interest waned. Funding for AI research decreased, leading to a period of reduced optimism and limited advancements.

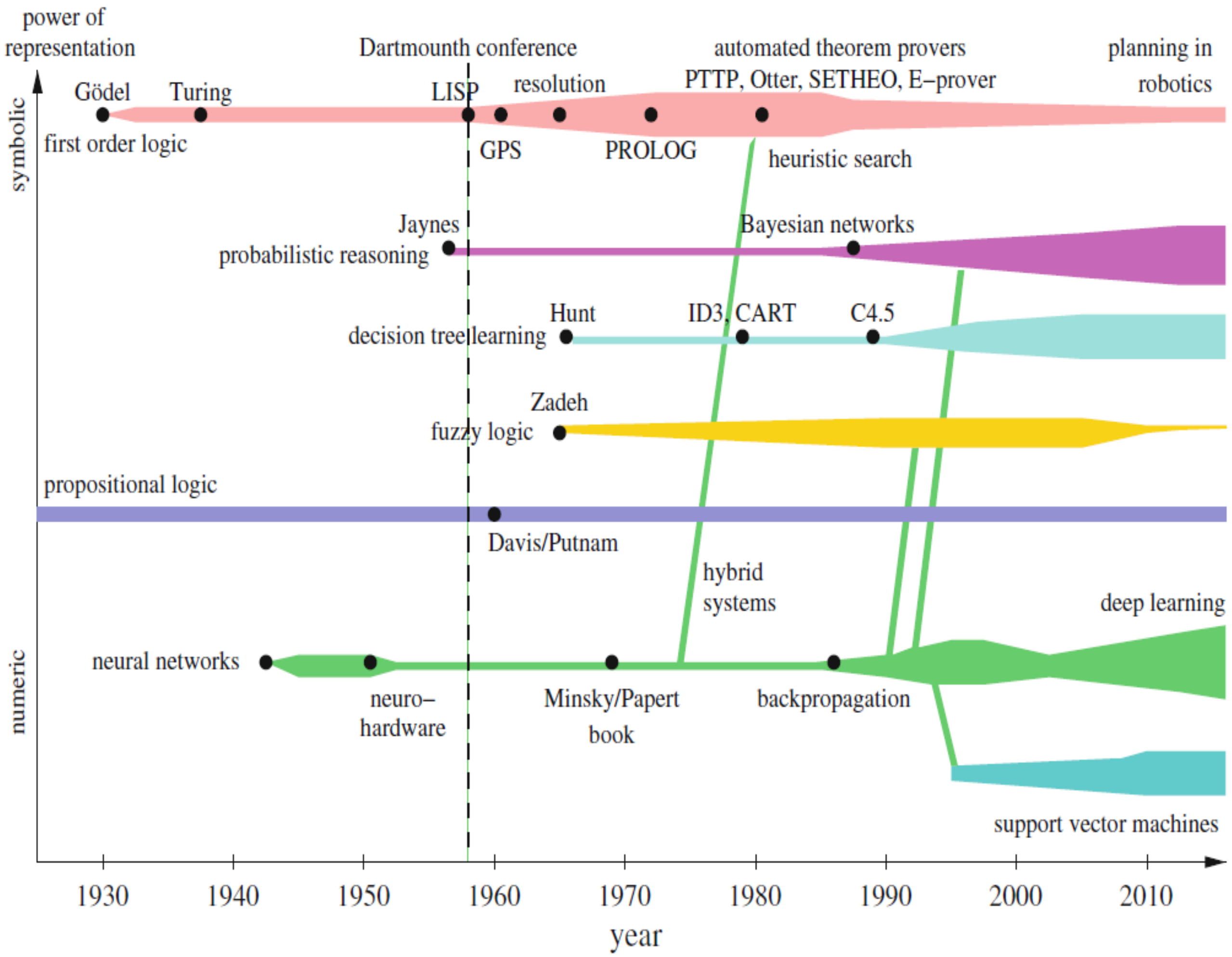



Technologies and Applications
Intelligent Agents
Intelligence; Tutoring
Natural Language Processing
Autonomous Vehicles
Personal Assistant
Voice Recognition
Speech Understanding
Robotics
Machine Learning
Automatic Programming
Neutral Networks
Genetic Algorithms
Game Playing
Deep Learning
Smart Cities
Computer Vision
Expert Systems
Smart Factories
Robo Advisors
Augmented Reality
Smart Homes
The AI Tree
Foundations
Philosophy
Mathematics
Computer Science
Human Behavior
Engineering
Neurology
IoT
M2M
Management Science
Logic
Information Systems
Sociology
Statistics
Psychology
Fuzzy Logic
Human Cognition
Pattern Recognition
Linguistics
Biology

The AI Tree

AI Techniques

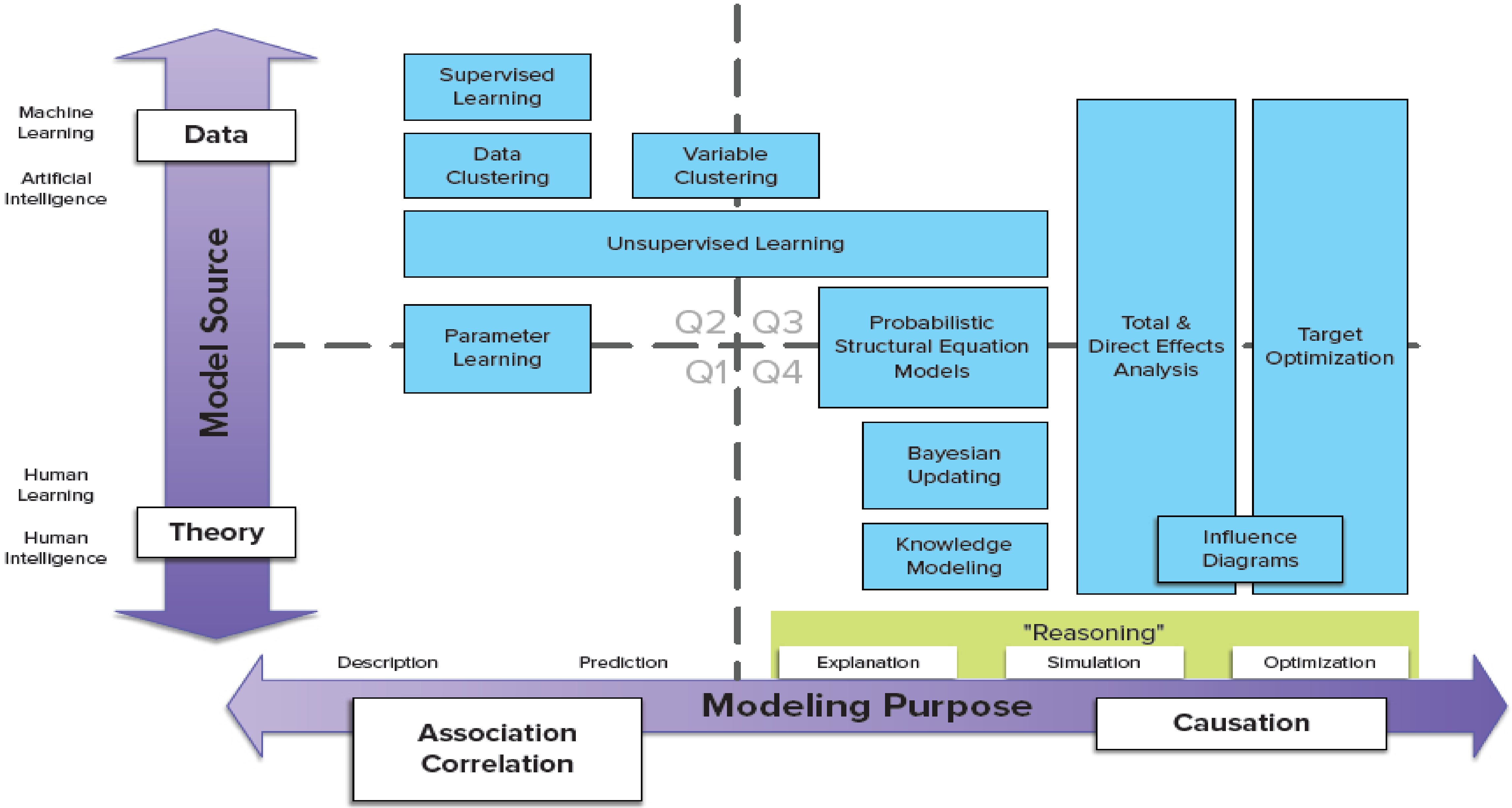




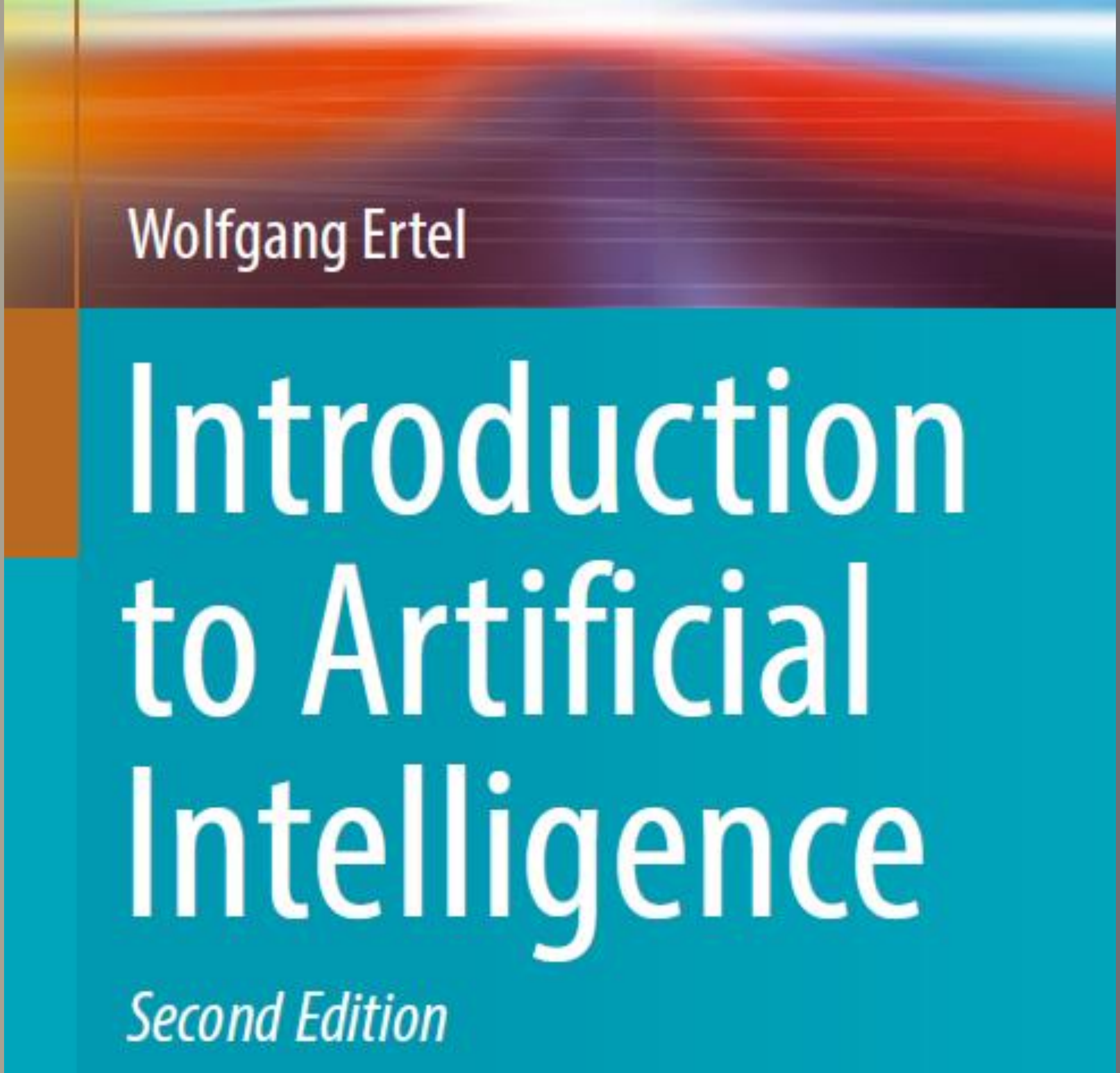
History of the various AI techniques. The width of the bars indicates prevalence of the techniques use.

Ref. [4]

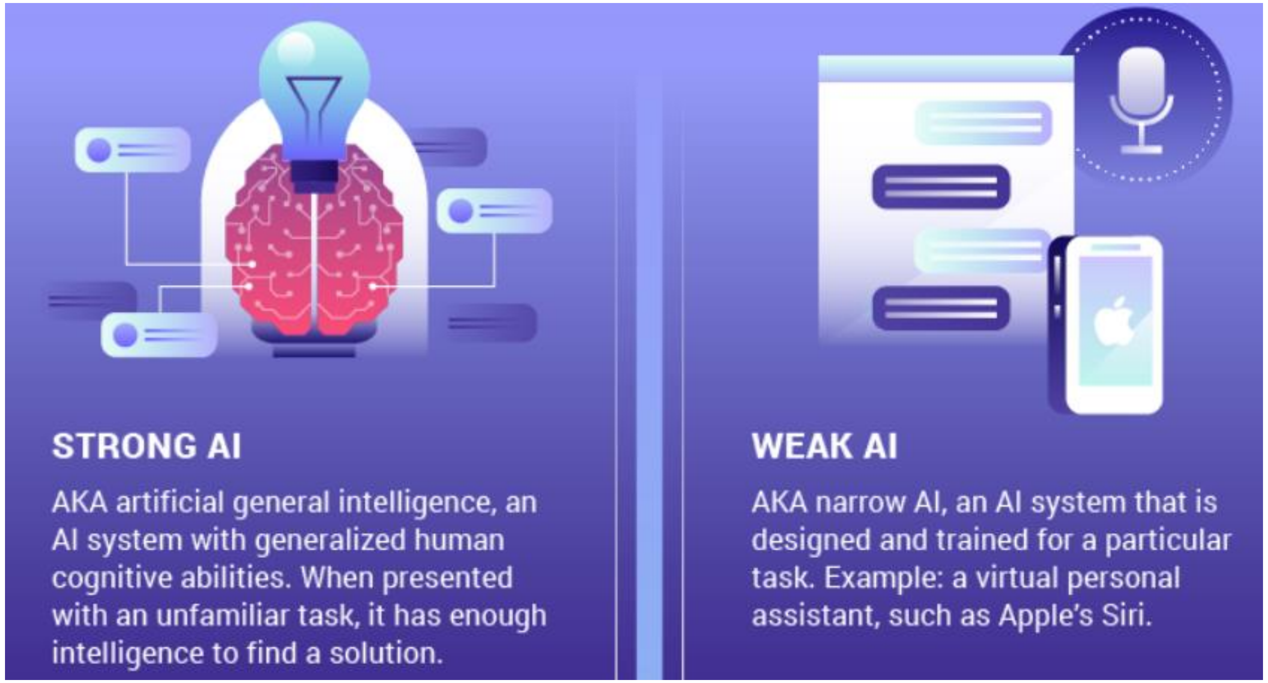
AI Techniques: Modelling Domain



Recommended Reading

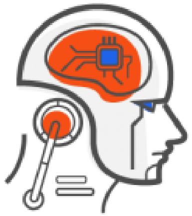


AI Classification



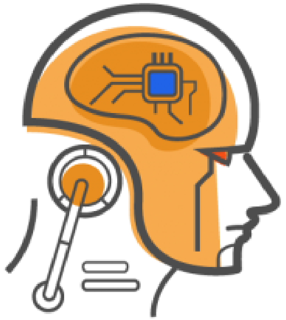
Narrow AI

Dedicated to assist with or take over specific tasks.



General AI

Takes knowledge from one domain, transfers to other domain.



Super AI

Machines that are an order of magnitude smarter than humans.

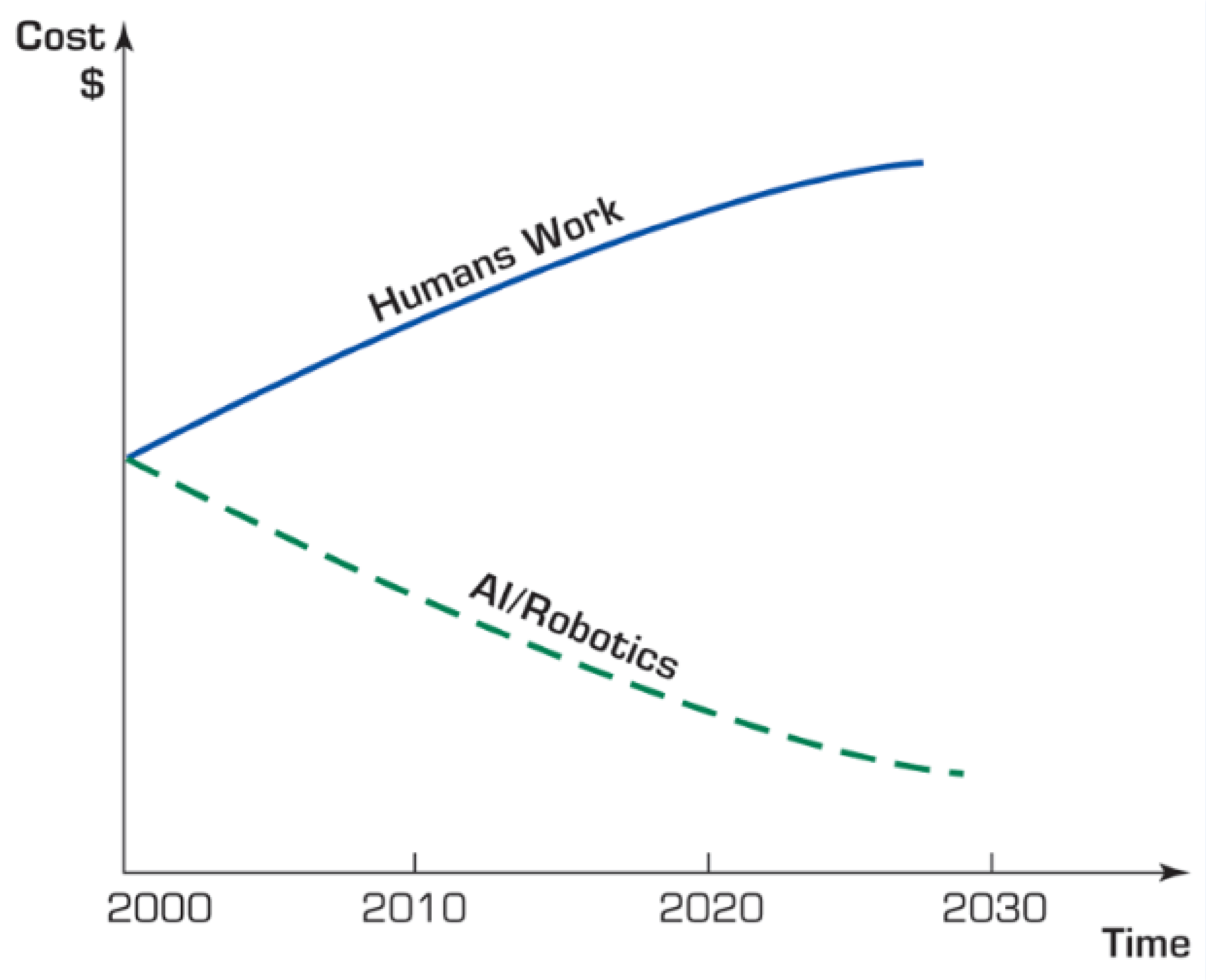
AI Claim

“You insist that there is something a machine cannot do. If you will tell me *precisely* what it is that a machine cannot do, then I can always make a machine which will do just that”

John von Neumann, 1948

- **Speed of execution** – While one doctor can make a diagnosis in ~10 minutes, AI system can make a million for the same time.
- **Less Biased** – They do not involve Biased opinions on decision making process.
- **Operational Ability** – They do not expect halt in their work due to saturation.
- **Accuracy** – Preciseness of the output obviously increases.

Cost Comparison AI vs. Human



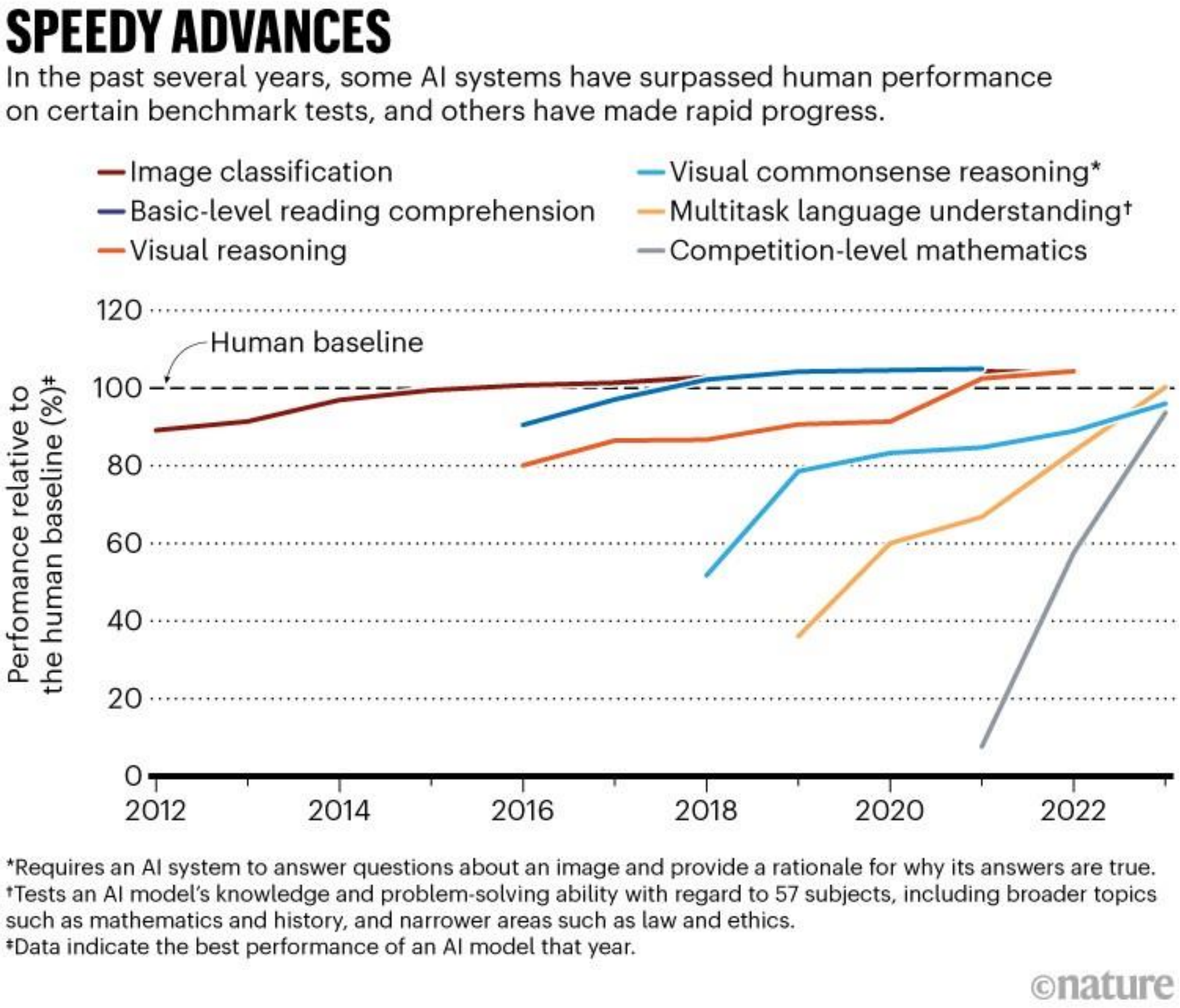
11 Times AI Beat Humans at Games, Art, Law and Everything in Between

AI is simply better than humans at lots of tasks, from the obvious of data analysis to the unexpected of music making.

- The Deep Blue computer is a better chess-player since 1997
- Question and Answer Computer Watson took down Jeopardy Champion, 2011
- AI is faster and smarter at visual recognition
- An AI music program writes better pop songs: Flow Machines
- AI system is more performant at strategy games too, 2016
- AI creates and critics original art
- AI just smashed the record for solving the Rubik's Cube: 0.38 secs

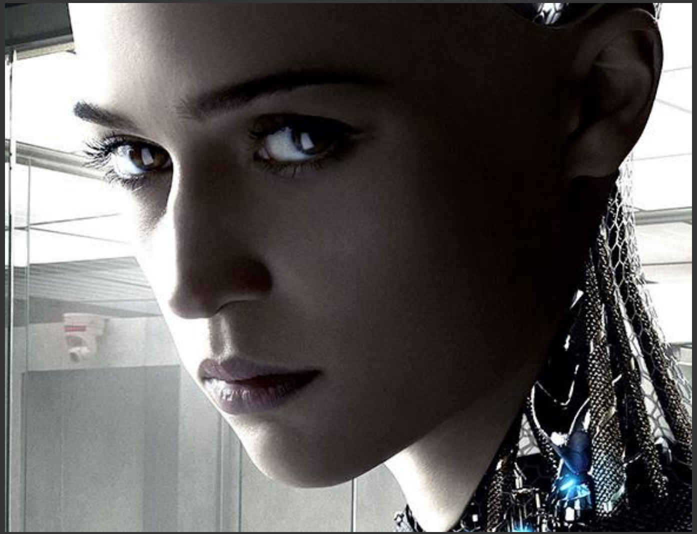
Ref. [5]

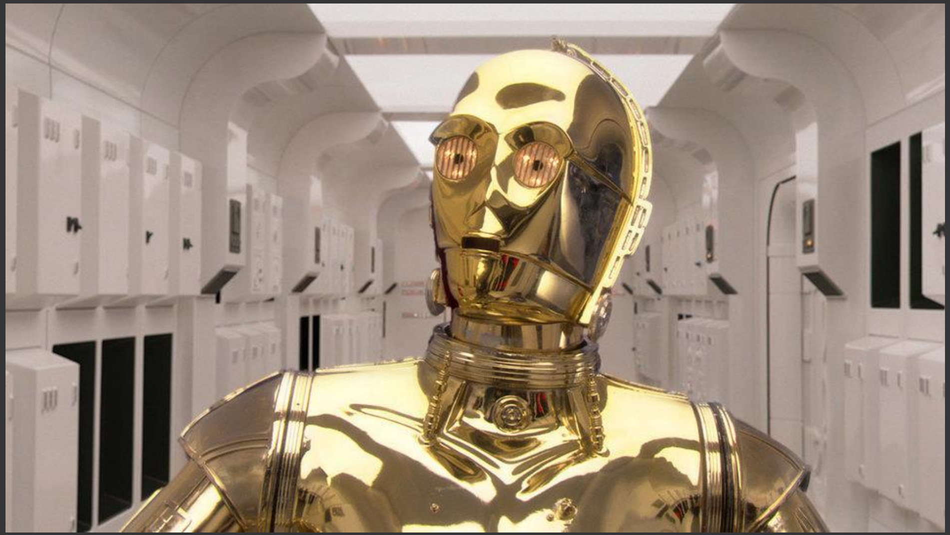

AI vs Humans:
Where AI Outperformed Us
From games to medicine, here's where artificial intelligence leads the way.
Games
AlphaGo beat world Go champion (2016)
OpenAI Five beat Dota 2 pros (2019)
Libratus crushed poker players (2017)
Knowledge & reasoning
Watson won Jeopardy (2011)
GPT-4 passes bar and SAT exams
Project Debater argues with humans on complex topics
Vision & health
AI radiologists outperform in diagnosing pneumonia, tumors
Self-driving cars detect obstacles faster than humans
AlphaFold solved protein folding challenge
Creativity
Flow Machines composes pop songs
DALL·E / Midjourney generate realistic art


SPEEDY ADVANCES
In the past several years, some AI systems have surpassed human performance on certain benchmark tests, and others have made rapid progress.
Image classification
Basic-level reading comprehension
Visual reasoning
Visual commonsense reasoning*
Multitask language understanding†
Competition-level mathematics
Human baseline
Perfomance relative to the human baseline (%)‡
120
100
80
60
40
20
0
2012
2014
2016
2018
2020
2022
*Requires an AI system to answer questions about an image and provide a rationale for why its answers are true.
†Tests an AI model's knowledge and problem-solving ability with regard to 57 subjects, including broader topics such as mathematics and history, and narrower areas such as law and ethics.
‡Data indicate the best performance of an AI model that year.
©nature

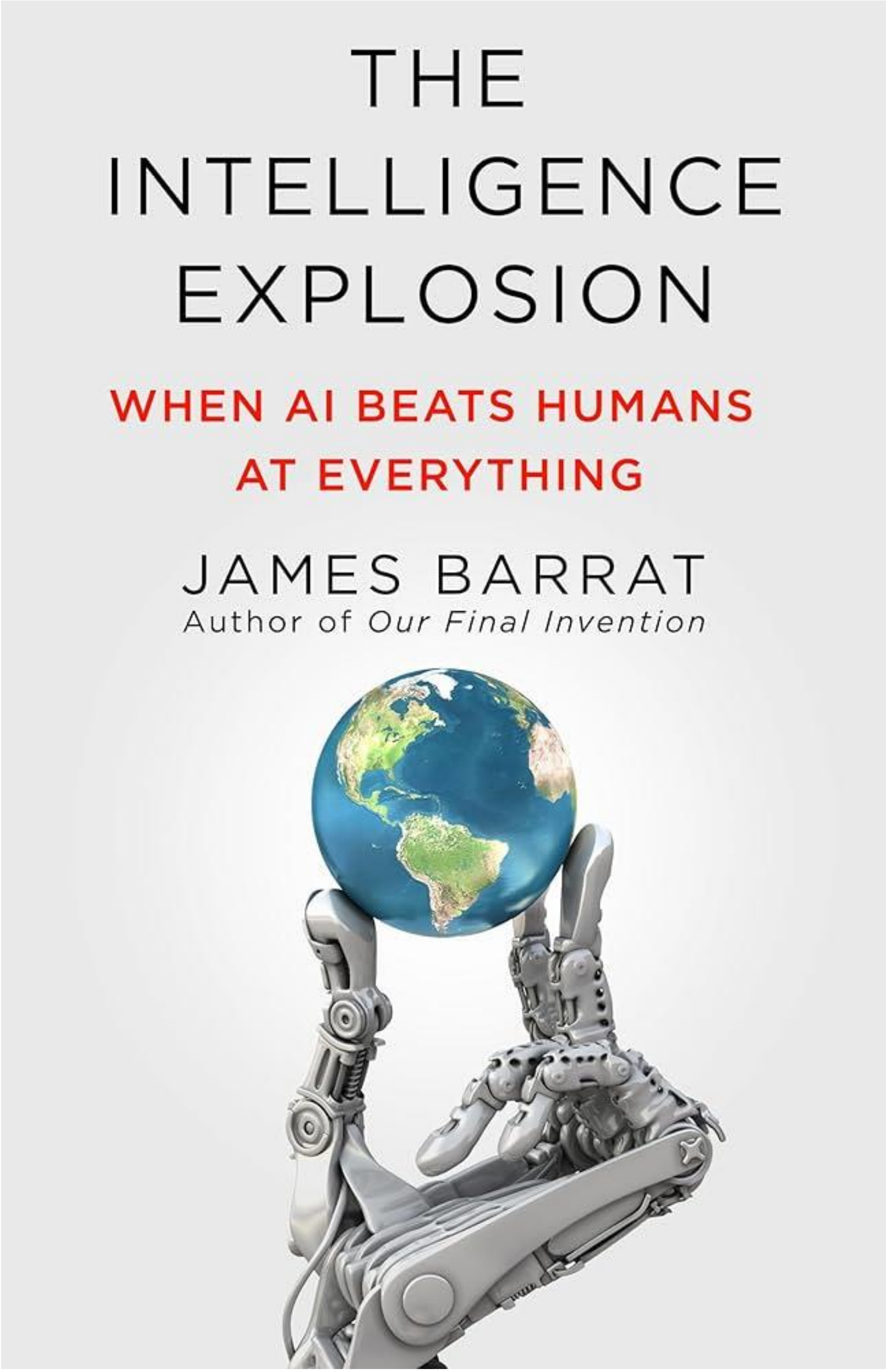
Ref. [6]

AI Isn't Exactly What we Thought it Was Going to be





Or Is It Going to be????

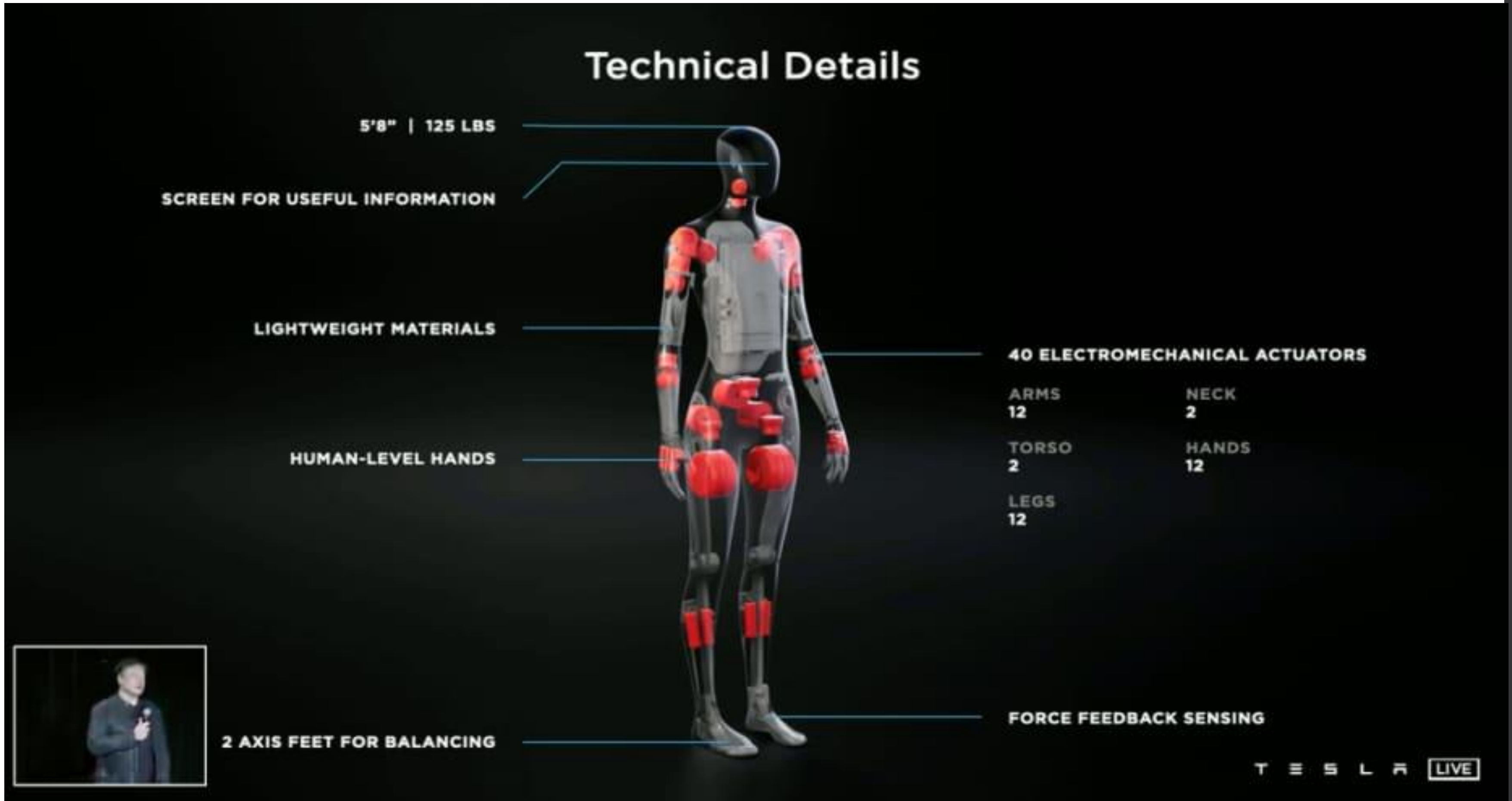


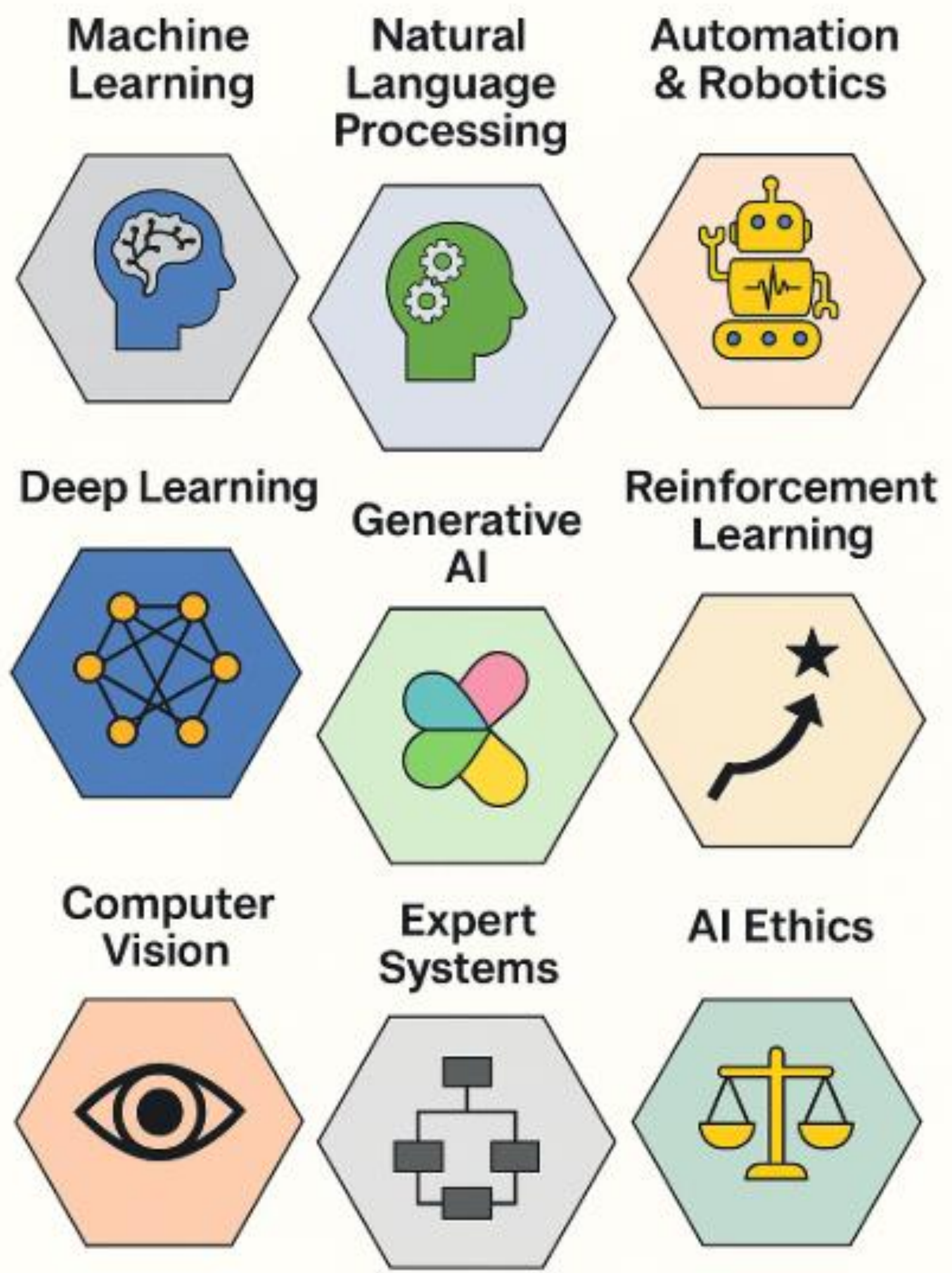
Tesla Bot

TECH

Elon Musk says Tesla will build a humanoid robot prototype by next year

PUBLISHED THU, AUG 19 2021•10:56 PM EDT | UPDATED FRI, AUG 20 2021•11:13 AM EDT





Top AI Techniques

AI vs. ML vs. DL

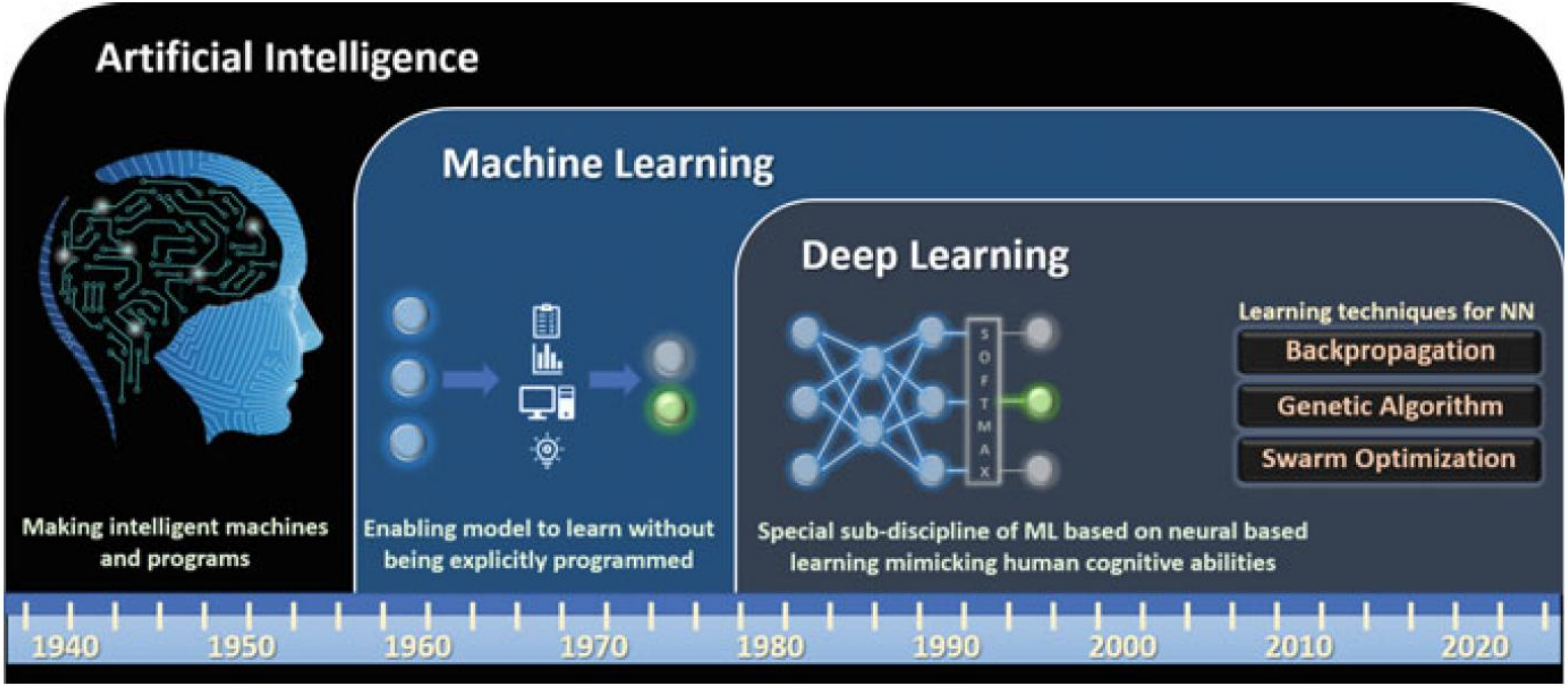


Image Source: Somani et al., Interpretability in Deep Learning

Reference

[1] https://www.kidscodecs.com/what-is-artificial-intelligence/

[2] https://techswizard.com/gadget/artificial-intelligence-vs-human-intelligence/

[3] https://www.educba.com/artificial-intelligence-vs-human-intelligence/

[4] Introduction to Artificial Intelligence, Wolfong Ertel, 2017.

[5] https://towardsdatascience.com/hey-siri-what-does-it-mean-to-be-human-f06fa6e251ce

[6] https://www.nature.com/articles/d41586-024-01087-4

Check Your Knowledge

Link

Thank You for Attention